RAJALAKSHMI ENGINEERING COLLEGE
(Autonomous Institution Affiliated to Anna University)
Department of Computer Science and Engineering

CS23512 Fundamentals of Mobile Computing

Class: III Year, 5th Sem CSE

UNIT 1

Topic

Introduction to Mobile Computing

Prepared by Dr. E.M.Malathy
Credits : Course Material IIT Bombay

Outcome Based Education: Course Outcomes

Course Outcomes: On completion of course, students will be able to
• Discover the characteristics of mobile computing applications including the major system components
• To explore the working model and characteristics of mobile computing
• To propose solutions with comparisons for problems related to mobile computing system through system investigations
• To identify the use of wireless technologies in appropriate applications and Quantum programming frameworks for mobile applications
• Develop a mobile application using mobile technologies and Emerging research level understanding at the intersection of mobile and quantum computing

Challenges

- Frequencies for Radio Transmission
- Signals
  - The representation for Mobile Communication
  - The propagation ranges.
- Antennas
  - Isotropic Radiation
  - Directive
- Signal propagation
  - Range
  - Multi path propagation.

Mobile Computing Today's Applications

- Wireless Voice/Data Communications
- Global Positioning Systems – Routing services
- Location Aware Services – Context-based
- Access to Remote Information Systems

Introduction

Goal of Wireless and Mobile Computing

"People and their machines should be able to access information and communicate with each other easily and securely, in any medium or combination of media –voice, data, image, video, or multimedia –anytime, anywhere, in a timely, cost-effective way."

Dr. G. H. Heilmeier, Oct 1992

Introduction to Wireless Networking Technologies

WLAN, WiFi Mesh and WiMAX

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Wireless networks

- Access computing/communication services, on the move

- Wireless WANs
  - Cellular Networks: GSM, GPRS, CDMA
  - Satellite Networks: Iridium

- Wireless LANs
  - WiFi Networks: 802.11
  - Personal Area Networks: Bluetooth

- Wireless MANs
  - WiMaX Networks: 802.16
  - Mesh Networks: Multi-hop WiFi
  - Adhoc Networks: useful when infrastructure not available

Limitations of the mobile environment

- Limitations of the Wireless Network
  - limited communication bandwidth
  - frequent disconnections
  - heterogeneity of fragmented networks
- Limitations Imposed by Mobility
  - route breakages
  - lack of mobility awareness by system/applications
- Limitations of the Mobile Device
  - short battery lifetime
  - limited capacities

Mobile communication

- Wireless vs. mobile Examples

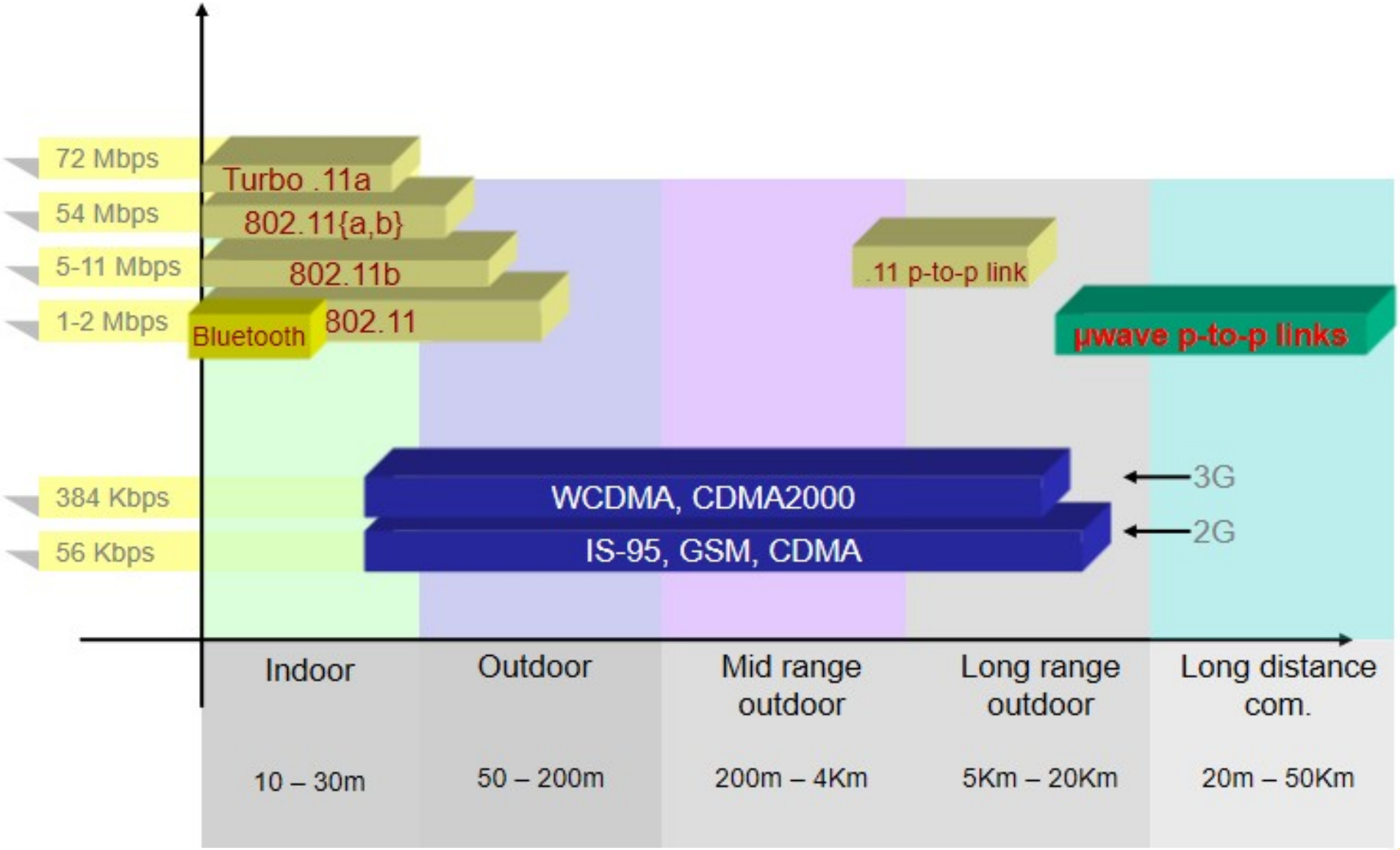
✗	✗	stationary computer
✗	✓	laptop in a hotel (portable)
✓	✗	wireless LAN in historic buildings
✓	✓	Personal Digital Assistant (PDA)

- Integration of wireless into existing fixed networks:
  - Local area networks: IEEE 802.11, ETSI (HIPERLAN)
  - Wide area networks: Cellular 3G, IEEE 802.16
  - Internet: Mobile IP extension

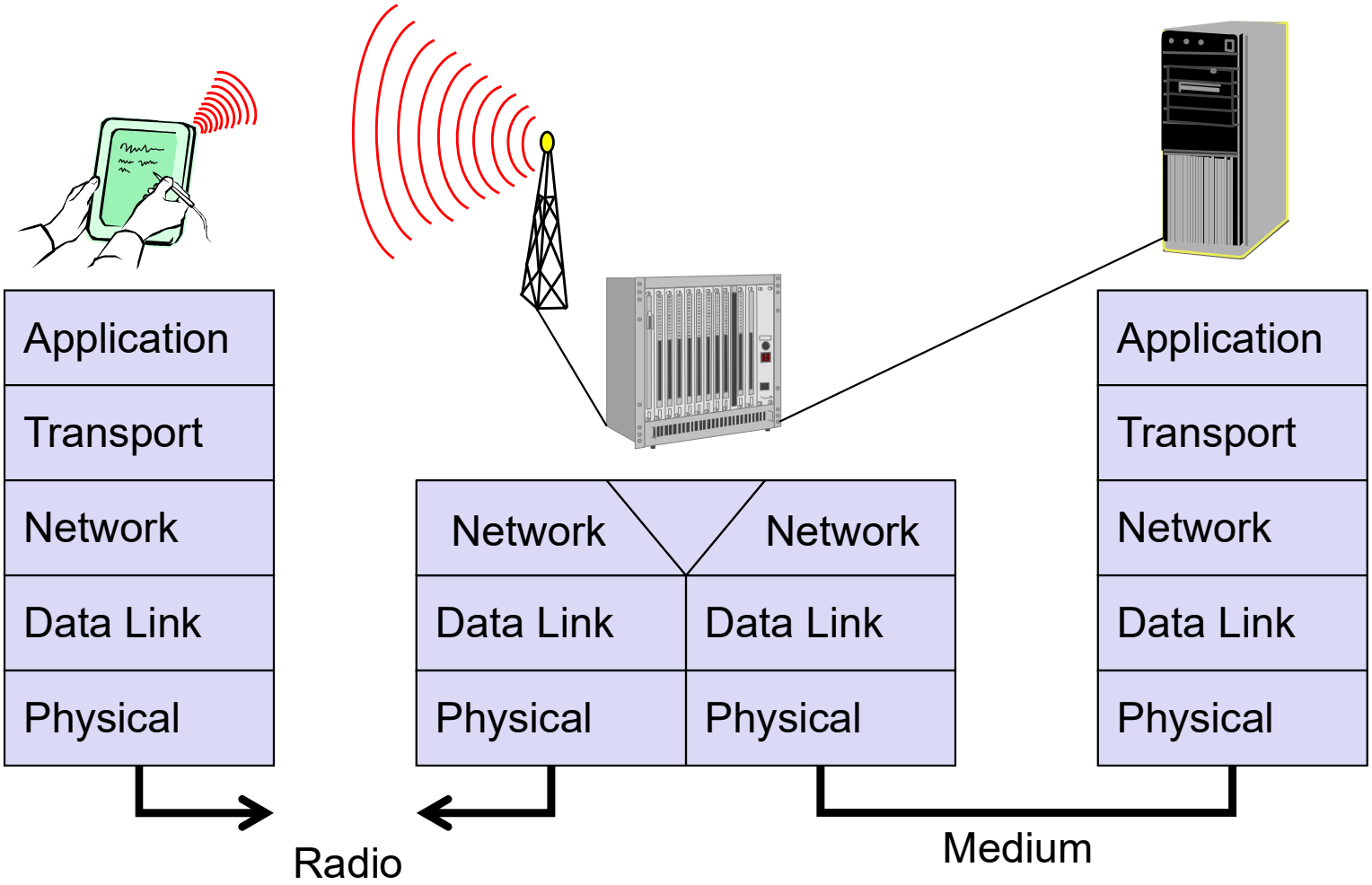
Wireless v/s Wired networks

- Regulations of frequencies
  - Limited availability, coordination is required
  - useful frequencies are almost all occupied
- Bandwidth and delays
  - Low transmission rates
    - few Kbits/s to some Mbit/s.
  - Higher delays
    - several hundred milliseconds
  - Higher loss rates
    - susceptible to interference, e.g., engines, lightning
- Always shared medium
  - Lower security, simpler active attacking
  - radio interface accessible for everyone
  - secure access mechanisms important

Wireless Technology Landscape



Reference model



Effect of mobility on protocol stack

- Application
  - new applications and adaptations
  - service location, multimedia
- Transport
  - congestion and flow control
  - quality of service
- Network
  - addressing and routing
  - device location, hand-over
- Link
  - media access and security
- Physical
  - transmission errors and interference

Perspectives

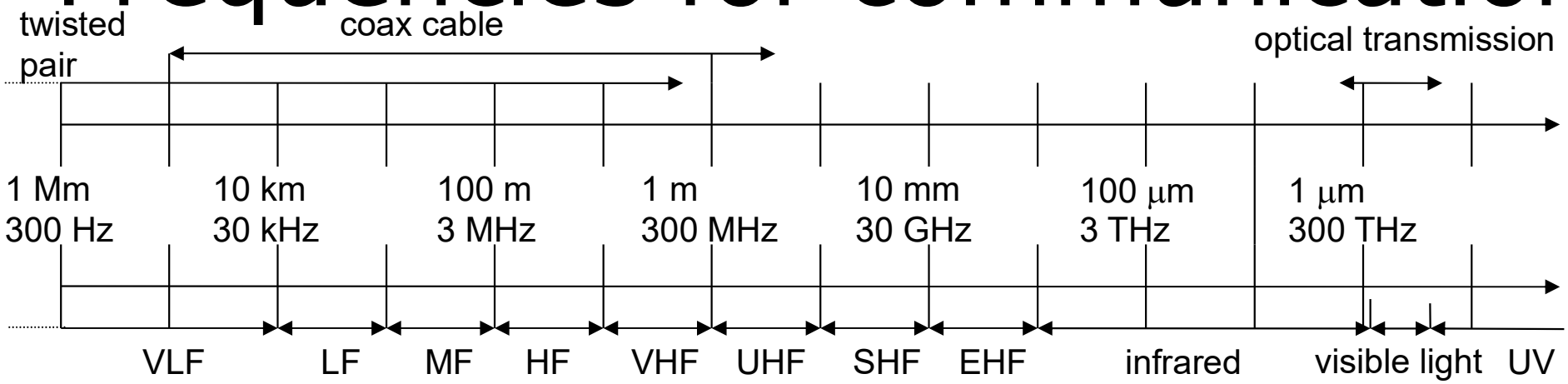
- Network designers: Concerned with cost-effective design
  - Need to ensure that network resources are efficiently utilized and fairly allocated to different users.
- Network users: Concerned with application services
  - Need guarantees that each message sent will be delivered without error within a certain amount of time.
- Network providers: Concerned with system administration
  - Need mechanisms for security, management, fault-tolerance and accounting.

RF Basics

Factors affecting wireless system design

- Frequency allocations
  - What range to operate? May need licenses.
- Multiple access mechanism
  - How do users share the medium without interfering?
- Antennas and propagation
  - What distances? Possible channel errors introduced.
- Signals encoding
  - How to improve the data rate?
- Error correction

Frequencies for communication



- VLF = Very Low Frequency
- LF = Low Frequency
- MF = Medium Frequency
- HF = High Frequency
- VHF = Very High Frequency
- UHF = Ultra High Frequency
- SHF = Super High Frequency
- EHF = Extra High Frequency
- UV = Ultraviolet Light

- Frequency and wave length: $\lambda = c/f$
- wave length λ, speed of light $c \cong 3x10^8 m/s$, frequency f

Wireless frequency allocation

- Radio frequencies range from 9KHz to 400GHZ (ITU)

- Microwave frequency range
  - 1 GHz to 40 GHz
  - Directional beams possible
  - Suitable for point-to-point transmission
  - Used for satellite communications
- Radio frequency range
  - 30 MHz to 1 GHz
  - Suitable for omnidirectional applications
- Infrared frequency range
  - Roughly, $3x10^{11}$ to $2x10^{14}$ Hz
  - Useful in local point-to-point multipoint applications within confined areas

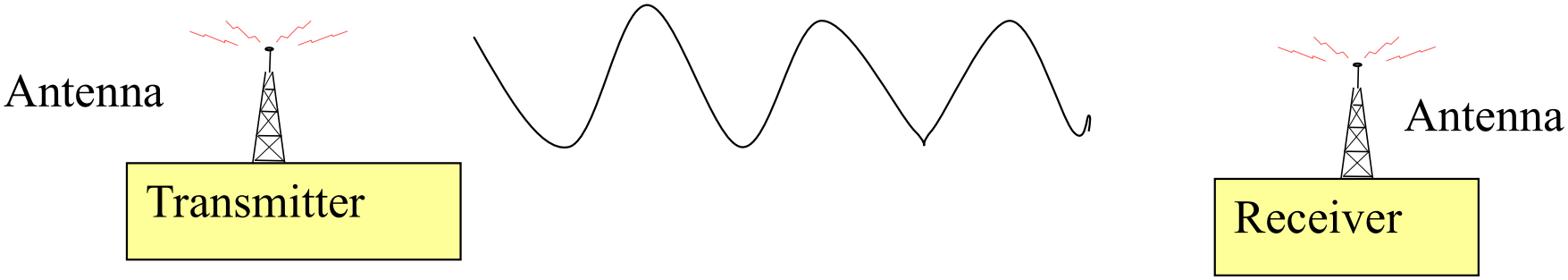
Frequencies for mobile communication

- VHF-/UHF-ranges for mobile radio
  - simple, small antenna for cars
  - deterministic propagation characteristics, reliable connections
- SHF and higher for directed radio links, satellite communication
  - small antenna, focusing
  - large bandwidth available
- Wireless LANs use frequencies in UHF to SHF spectrum
  - some systems planned up to EHF
  - limitations due to absorption by water and oxygen molecules (resonance frequencies)
    - weather dependent fading, signal loss caused by heavy rainfall etc.

Frequency regulations

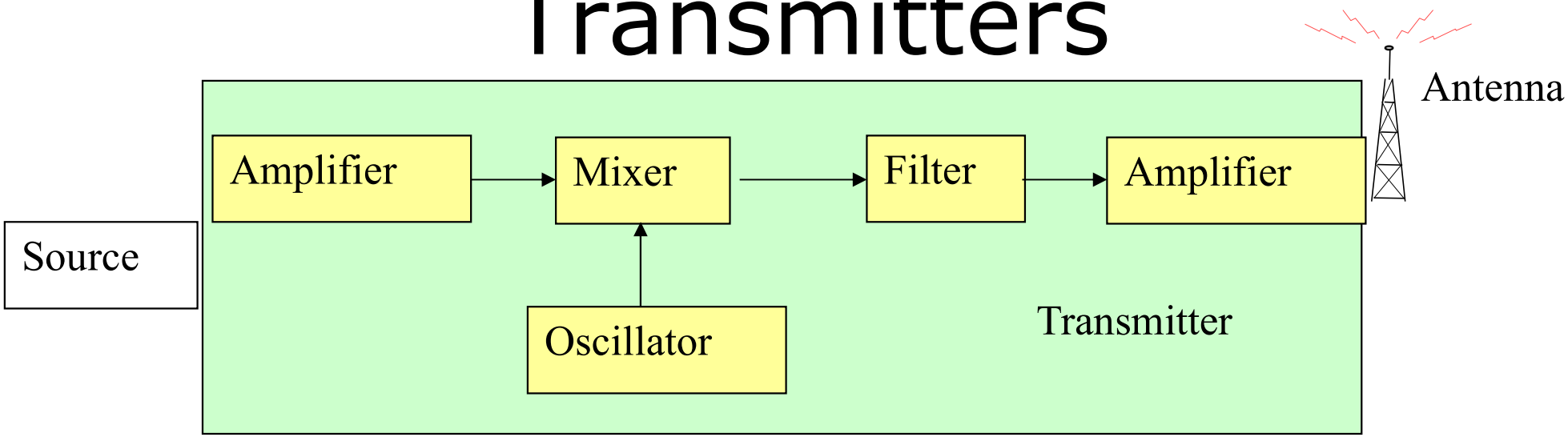
- Frequencies from 9KHz to 300 MHZ in high demand (especially VHF: 30-300MHZ)
- Two unlicensed bands
  - Industrial, Science, and Medicine (ISM): 2.4 GHz
  - Unlicensed National Information Infrastructure (UNII): 5.2 GHz
- Different agencies license and regulate
  - www.fcc.gov - US
  - www.etsi.org - Europe
  - www.wpc.dot.gov.in - India
  - www.itu.org - International co-ordination
- Regional, national, and international issues
- Procedures for military, emergency, air traffic control, etc

Wireless transmission



- Wireless communication systems consist of:
  - Transmitters
  - Antennas: radiates electromagnetic energy into air
  - Receivers
- In some cases, transmitters and receivers are on same device, called transceivers.

Transmitters



Suppose you want to generate a signal that is sent at 900 MHz and the original source generates a signal at 300 MHz.

- Amplifier - strengthens the initial signal
- Oscillator - creates a carrier wave of 600 MHz
- Mixer - combines signal with oscillator and produces 900 MHz (also does modulation, etc)
- Filter - selects correct frequency
- Amplifier - Strengthens the signal before sending it

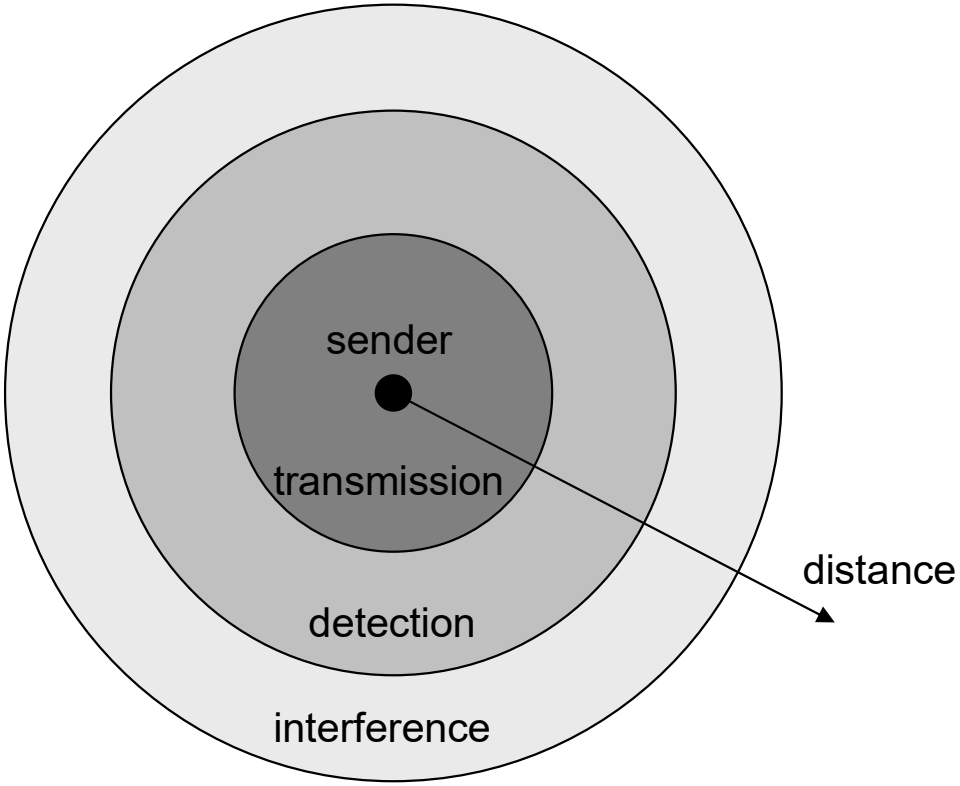
Signals

- physical representation of data
- function of time and location
- signal parameters: parameters representing the value of data
- classification
  - continuous time/discrete time
  - continuous values/discrete values
  - analog signal = continuous time and continuous values
  - digital signal = discrete time and discrete values
- signal parameters of periodic signals: period T, frequency f=1/T, amplitude A, phase shift φ
  - sine wave as special periodic signal for a carrier:

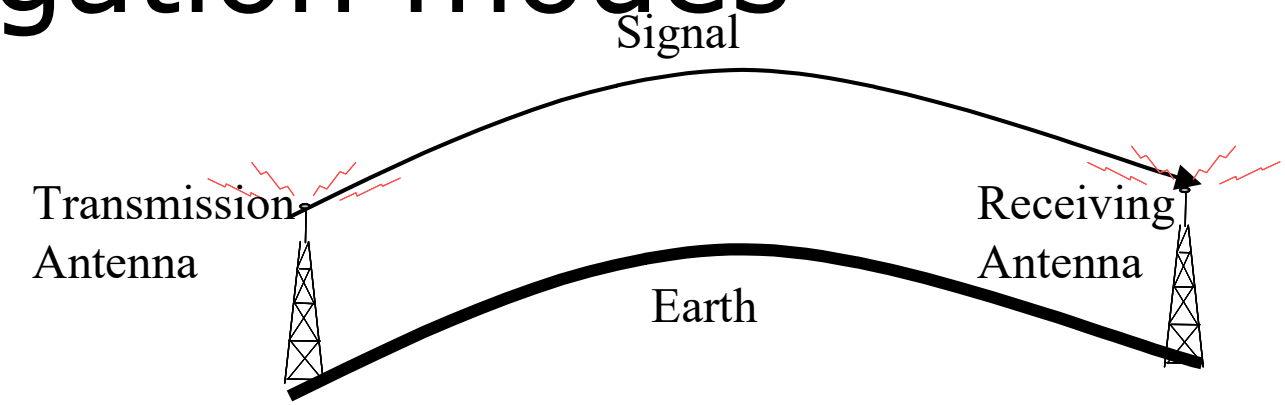
$$s(t) = A_t \sin(2 \pi f_t t + \varphi_t)$$

Signal propagation ranges

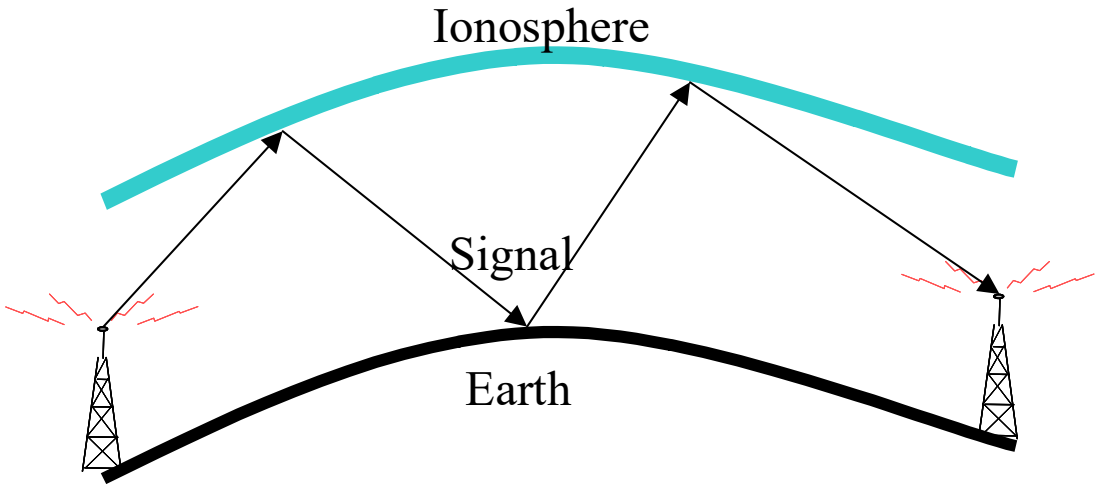
- Transmission range
  - communication possible
  - low error rate
- Detection range
  - detection of the signal possible
  - no communication possible
- Interference range
  - signal may not be detected
  - signal adds to the background noise



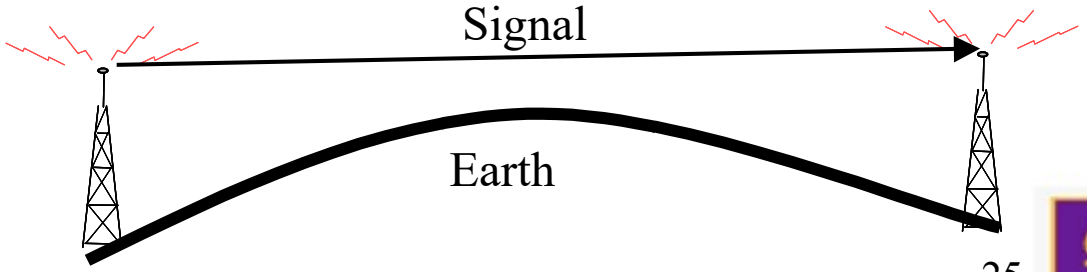
Propagation modes



a) Ground Wave Propagation



b) Sky Wave Propagation



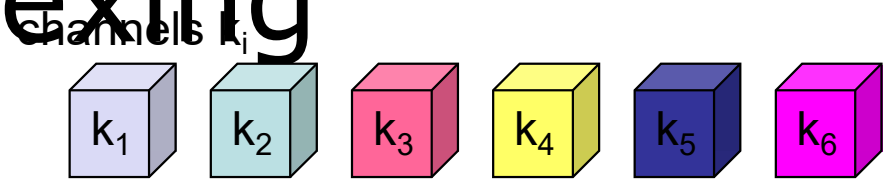
c) Line-of-Sight Propagation

Modulation

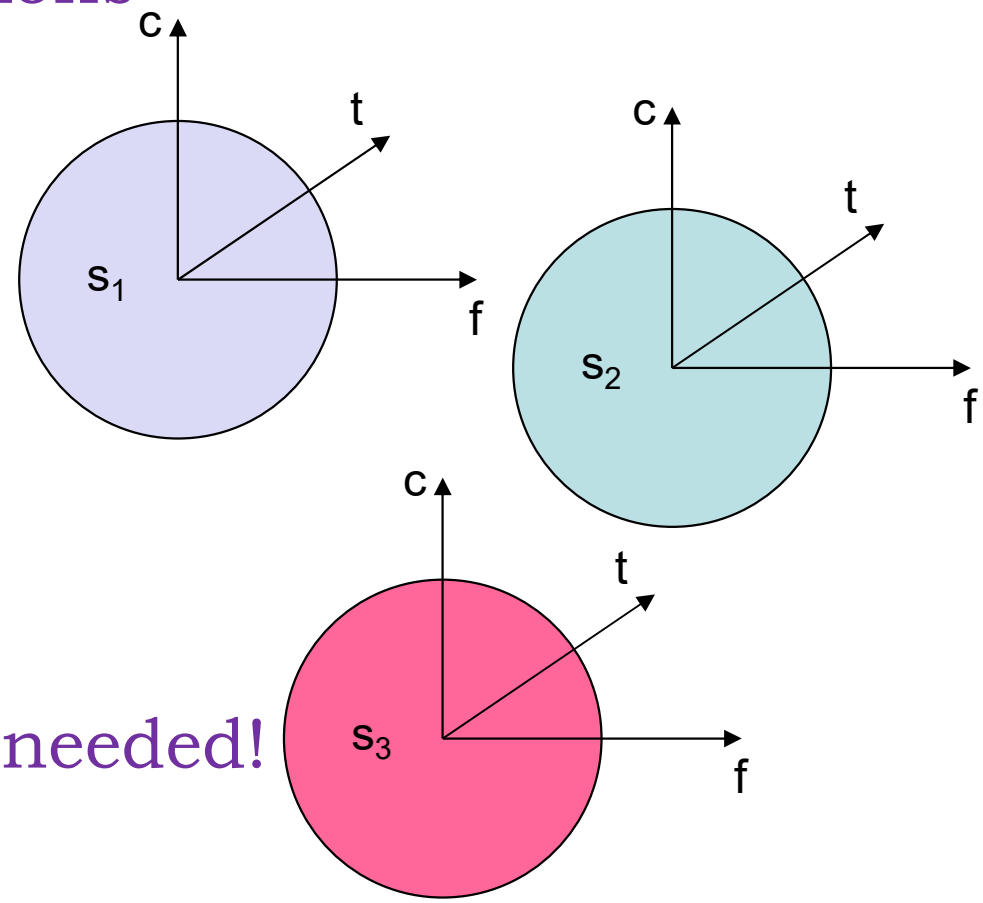
- Digital modulation
  - digital data is translated into an analog signal (baseband)
  - ASK, FSK, PSK
  - differences in spectral efficiency, power efficiency, robustness
- Analog modulation
  - shifts center frequency of baseband signal up to the radio carrier
- Motivation
  - smaller antennas (e.g., $\lambda/4$)
  - Frequency Division Multiplexing
  - medium characteristics
- Basic schemes
  - Amplitude Modulation (AM)
  - Frequency Modulation (FM)
  - Phase Modulation (PM)

Multiplexing Mechanisms

Multiplexing



- Multiplexing in 4 dimensions
  - space (s_i)
  - time (t)
  - frequency (f)
  - code (c)



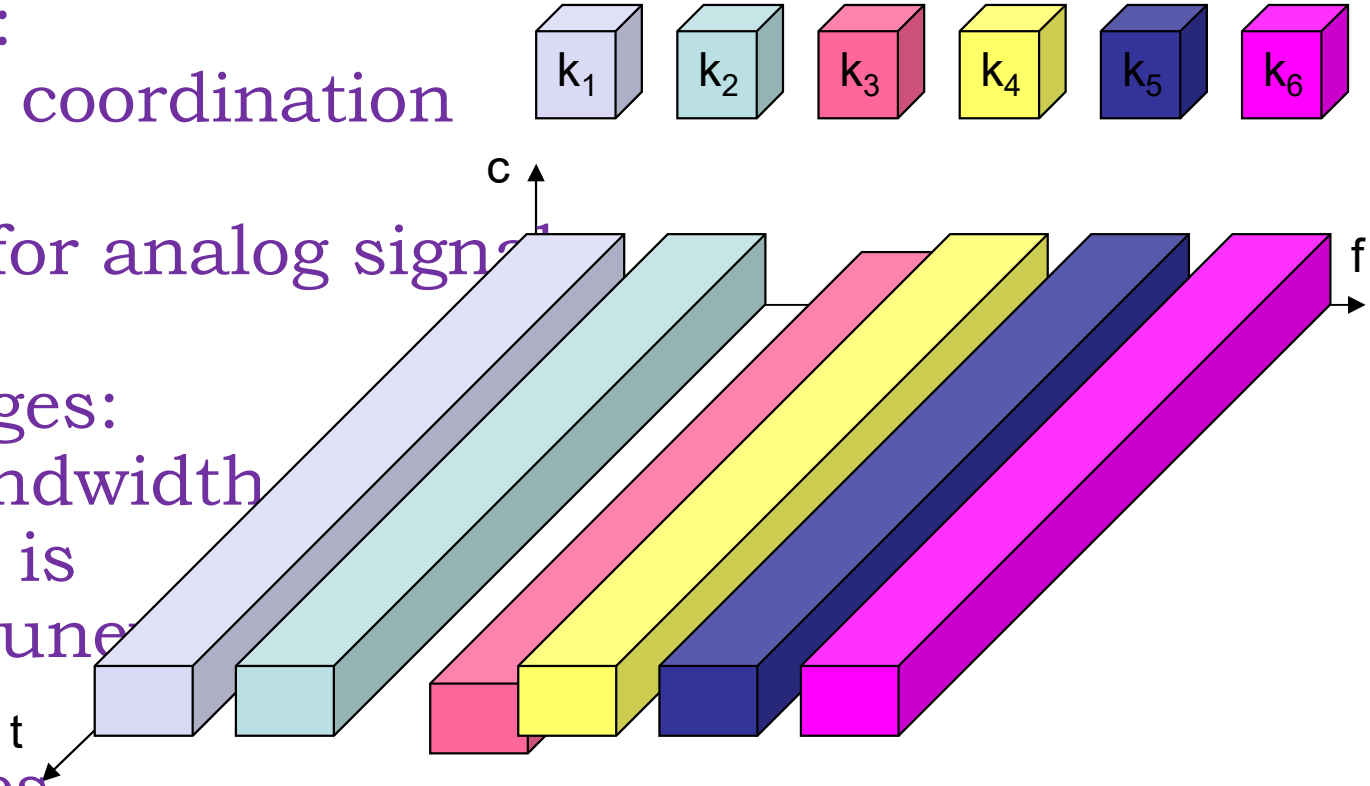
- Goal: multiple use of a shared medium

- Important: guard spaces needed!

Frequency multiplex

- Separation of the whole spectrum into smaller frequency bands
- A channel gets a certain band of the spectrum for the whole time
- Advantages:
- no dynamic coordination necessary
- works also for analog signal

- Disadvantages:
- waste of bandwidth if the traffic is distributed unev
- inflexible
- guard spaces

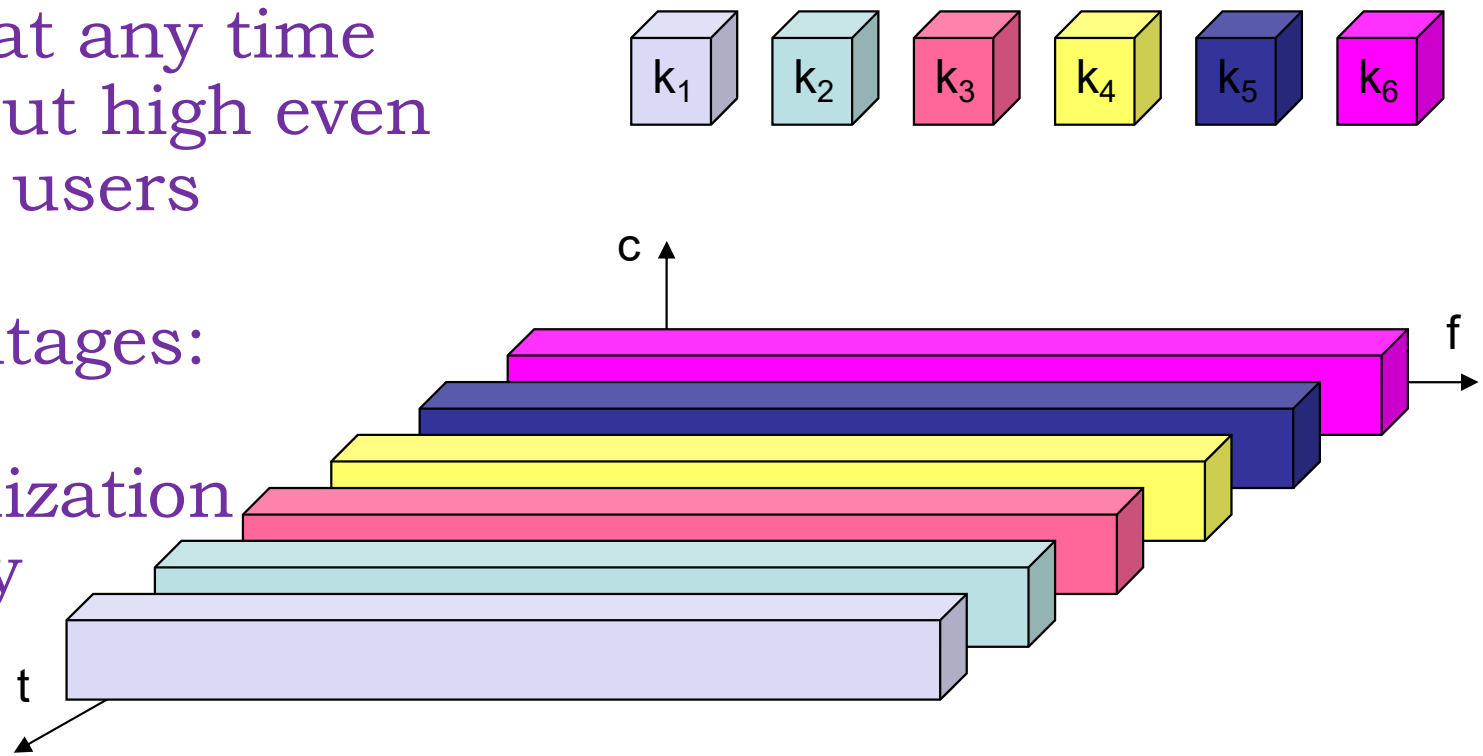


Time multiplex

- A channel gets the whole spectrum for a certain amount of time

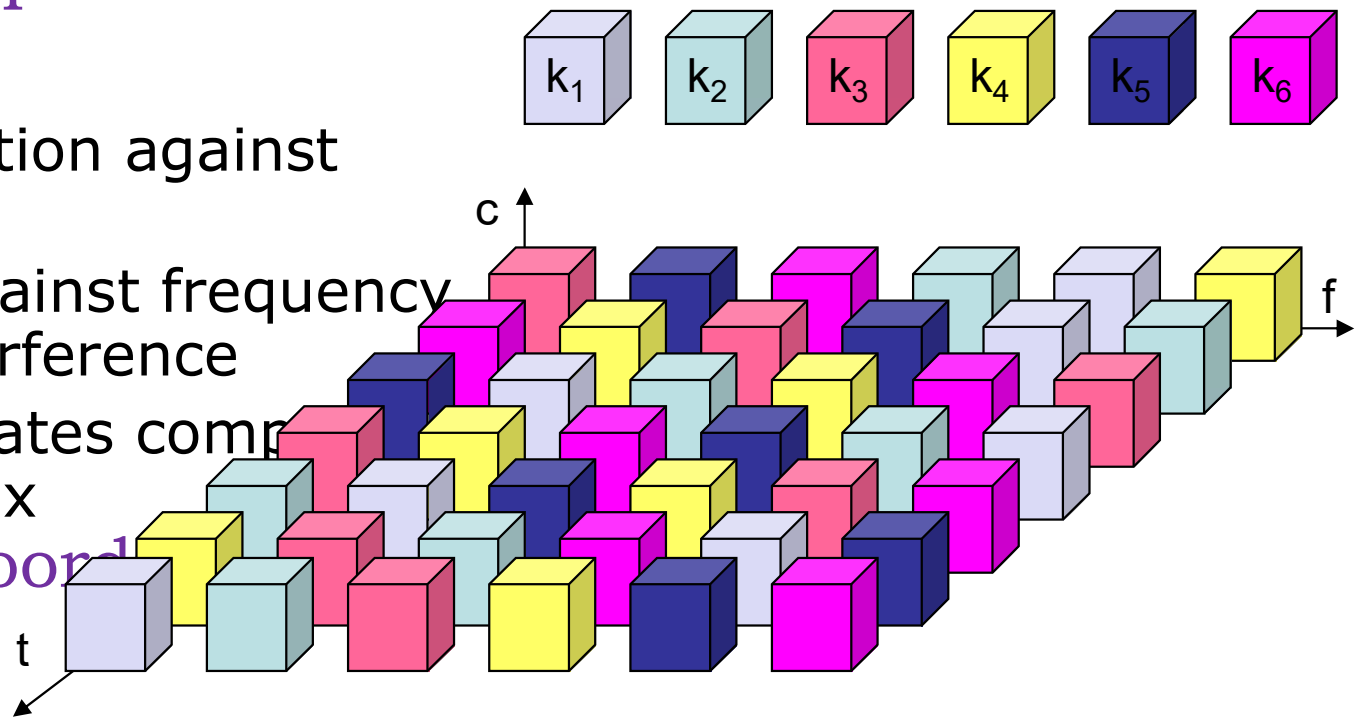
- Advantages:
- only one carrier in the medium at any time
- throughput high even for many users

- Disadvantages:
- precise synchronization necessary



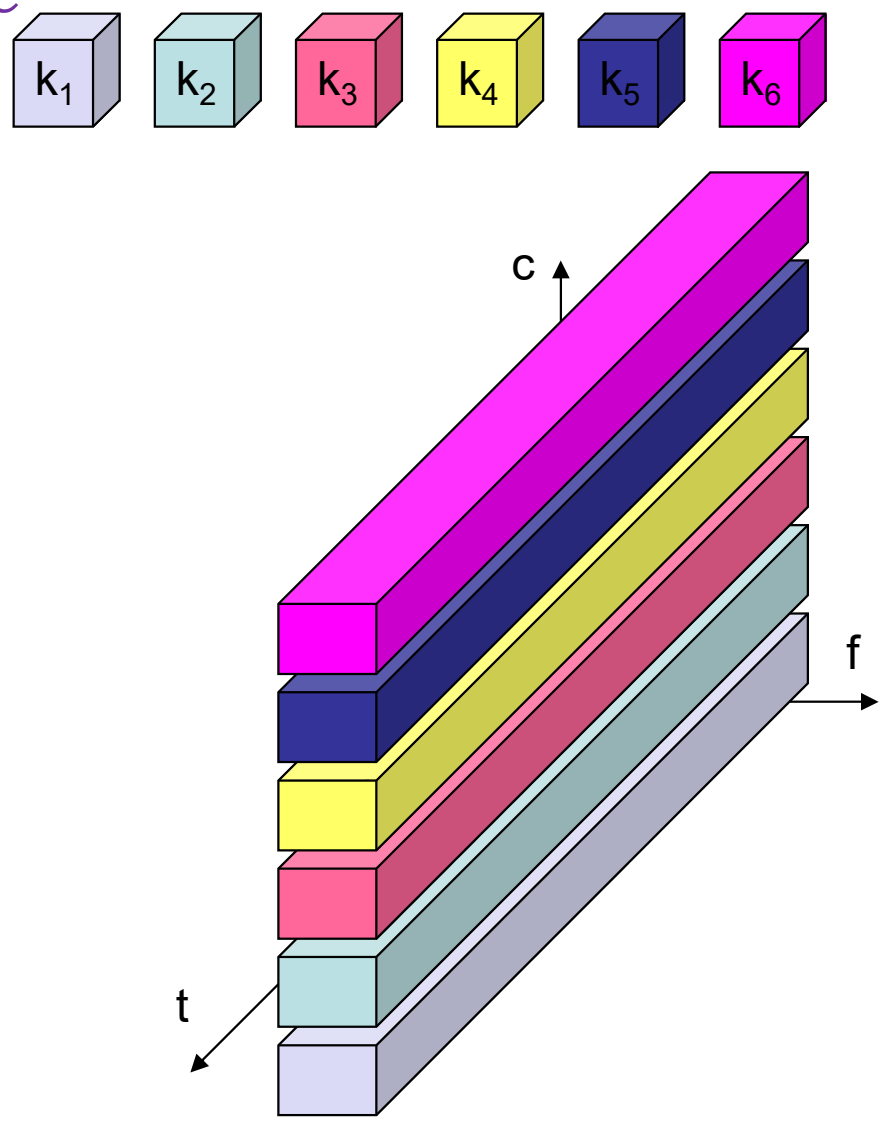
Time and frequency multiplex

- Combination of both methods
- A channel gets a certain frequency band for a certain amount of time
- Example: GSM
- Advantages:
  - better protection against tapping
  - protection against frequency selective interference
  - higher data rates comp code multiplex
- but: precise coord required



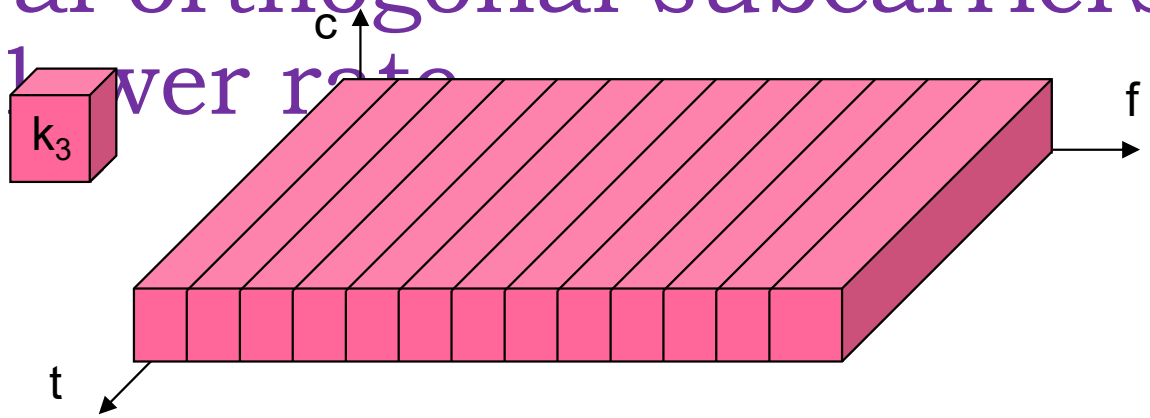
Code multiplex

- Each channel has a unique code
- All channels use the same spectrum at the same time
- Advantages:
  - bandwidth efficient
  - no coordination and synchronization necessary
  - good protection against interference and tapping
- Disadvantages:
  - lower user data rates
  - more complex signal regeneration
- Implemented using spread spectrum technology



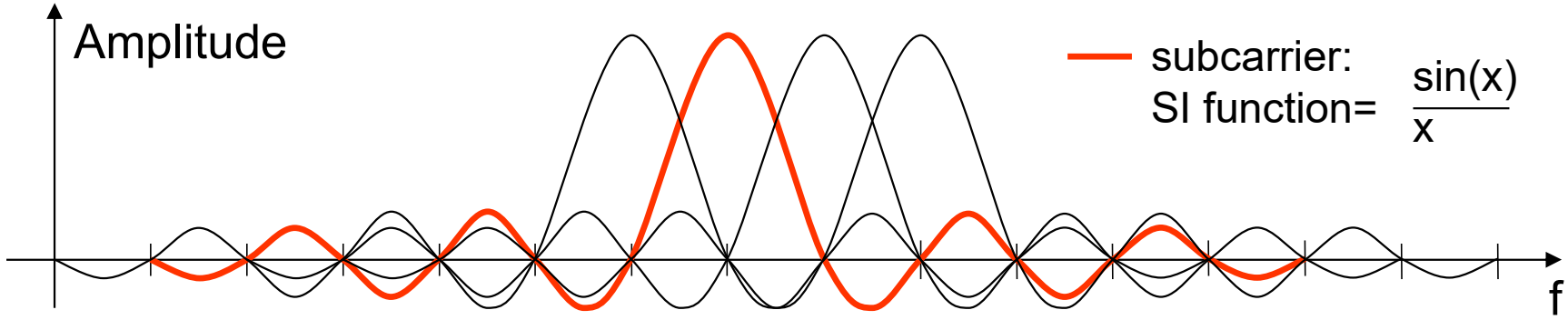
OFDM (Orthogonal Frequency Division)

- Parallel data transmission on several orthogonal subcarriers with lower rate



Maximum of one subcarrier frequency appears exactly at a frequency where all other subcarriers equal zero

- superposition of frequencies in the same frequency range



OFDM

- Properties
  - Lower data rate on each subcarrier ➔ less ISI
  - interference on one frequency results in interference of one subcarrier only
  - no guard space necessary
  - orthogonality allows for signal separation via inverse FFT on receiver side
  - precise synchronization necessary (sender/receiver)

- Advantages
  - no equalizer necessary
  - no expensive filters with sharp edges necessary
  - better spectral efficiency (compared to CDM)
- Application
  - 802.11a, HiperLAN2, ADSL

ALOHA

Stations transmit whenever they have data to send

- Detect collision or wait for acknowledgment
- If no acknowledgment (or collision), try again after a random waiting time

Collision: If more than one node transmits at the same time

If there is a collision, all nodes have to re-transmit packets

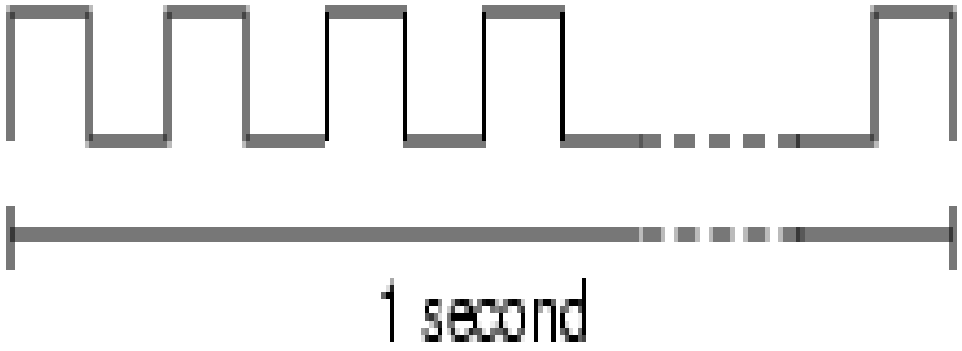
Bandwidth and Delay

Bandwidth

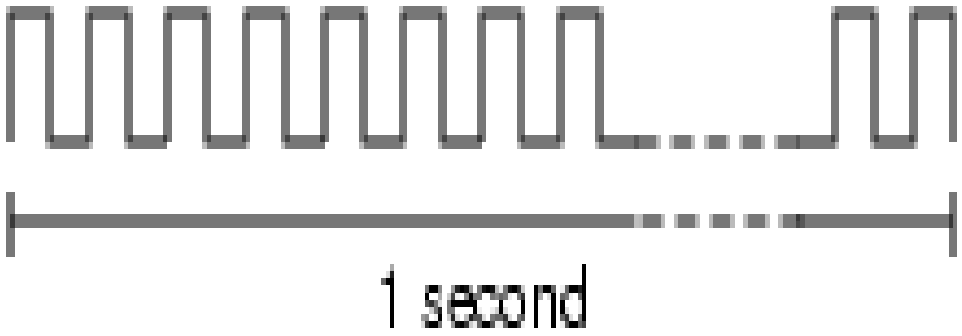
- Amount of data that can be transmitted per unit time
  - expressed in cycles per second, or Hertz (Hz) for analog devices
  - expressed in bits per second (bps) for digital devices
  - KB = 2^10 bytes; Mbps = 10^6 bps

- Link v/s End-to-End

Bandwidth v/s bit width



1 Mbps
(each bit 1 microseconds wide)



2 Mbps
(each bit 0.5 microseconds wide)

Latency (delay)

- Time it takes to send message from point A to point B
  - Latency = Propagation + Transmit + Queue
  - Propagation = Distance / SpeedOfLight
  - Transmit = Size / Bandwidth

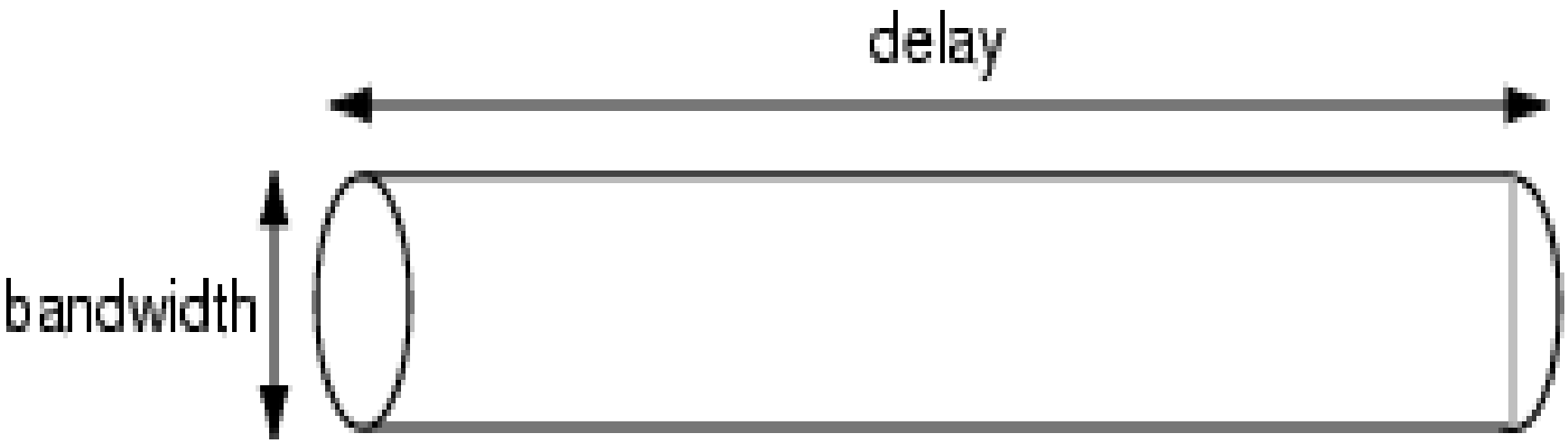
- Queueing not relevant for direct links
- Bandwidth not relevant if Size = 1 bit
- Software overhead can dominate when Distance is small

- RTT: round-trip time

Delay X Bandwidth product

- Relative importance of bandwidth and delay
- Small message: 1ms vs 100ms dominates 1Mbps vs 100Mbps
- Large message: 1Mbps vs 100Mbps dominates 1ms vs 100ms

Delay X Bandwidth product



100ms RTT and 45Mbps Bandwidth = 560 KB of data

TCP/IP Basics

Interconnection devices

Basic Idea: Transfer data from input to output

- Repeater
  - Amplifies the signal received on input and transmits it on output
- Switch
  - Reads destination address of each packet and forwards appropriately to specific port
  - Layer 3 switches (IP switches) also perform routing functions
- Router
  - decides routes for packets, based on destination address and network topology
  - Exchanges information with other routers to learn network topology

TCP/IP layers

- Physical Layer:
  - Transmitting bits over a channel.
  - Deals with electrical and procedural interface to the transmission medium.

- Data Link Layer:
  - Transform the raw physical layer into a `link' for the higher layer.
  - Deals with framing, error detection, correction and multiple access.

TCP/IP layers (contd.)

- Network Layer:
  - Addressing and routing of packets.
  - Deals with subnetting, route determination.

- Transport Layer:
  - end-to-end connection characteristics.
  - Deals with retransmissions, sequencing and congestion control.

TCP/IP layers (contd.)

- Application Layer:
  - ``application'' protocols.
  - Deals with providing services to users and application developers.

- Protocols are the building blocks of a network architecture.

Study of Technology 1G to 5G- Mobile computing

Class : III yr- 5 SEM- CSE

Course Instructor : Dr. E. M. Malathy

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Wireless Networking Generations

The generation characteristics include:

- **1G**:
  - ✓ Deployed in the 1980s
  - ✓ Based on analog FM transmission technologies.

- **2G**:
  - ✓ Introduced on market in 1991
  - ✓ Applied digital technologies such as TDMA, GSM, CDMA.
  - ✓ 2G systems offered higher capacity
  - ✓ Lower costs for network operators as well as short messages
  - ✓ Low-rate data services for users.

Wireless Networking Generations

❑ 2.5G

- ✓ *A technology* between *2G* and *3G* of mobile telephony.
- ✓ Described as *2G* cellular technology combined with *GPRS.*
- ✓ Introduced ability to use packet switched radio connections over the air.
- ✓ GPRS offers users the opportunity to always be online but only be charged for data transferred.

Wireless Networking Generations

❑3G

- ✓ Maximum bit rate of 2 Mbps .
- ✓ packet-switched multimedia services (data, video, etc.)
  as well as circuit-switched voice services.
- ✓ GSM/TDMA operators can evolve towards EDGE or WCDMA with higher and variable bit rates
  CDMA operators can evolve to cdma2000
- ✓ Improved spectrum efficiency.

Introduction to 4G

- ✓ Developed Objectives to overcome the shortcomings and limitations of 3G.
- ✓ To offer higher bandwidths up to value of 100Mbps.
- ✓ 4G promises voice, data and high-quality multimedia in real-time ("streamed") form all the time and anywhere.
- ✓ 4G are expected to consist of a collection of wireless networks

Includes

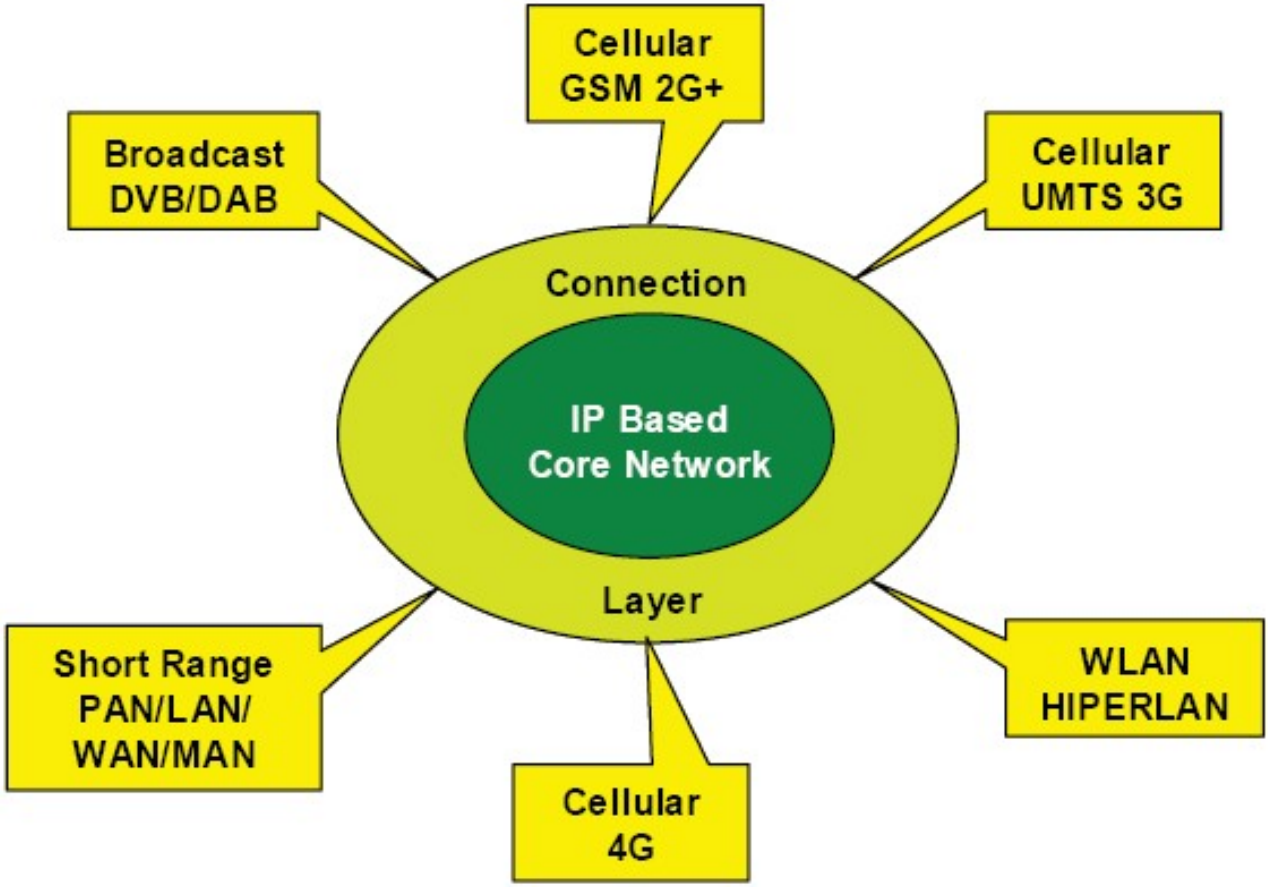
- Personal Area Network using, for example Bluetooth,
- Local area networks using WLAN,
- The satellite – based mobile networks.

COMPARISON BETWEEN 3G AND 4G

Technology	3G	4G
Frequency band	1.8 - 2.5 GHz	2 – 8 GHz
Bandwidth	5-20 MHz	5-20 MHz
Data rate	Up to 2Mbps	100Mbps moving 1GB stationary
Access	W-CDMA	VSF-OFCDM VSF-CDMA
FEC	Turbo-codes	Concatenated
Switching	Circuit/Packet	Packet

Features of 4G Networks

- ✓ 4G networks are all-IP (Internet Protocol) based heterogeneous networks
- ✓ This will allow users to:
  - Select any system at any time and any where
  - Use Multiple systems at the same time (e.g. GPS and WLANs and CDMA)
- ✓ A wide range of applications using only one 4G integrated terminal



4G Key challenge Mobile station

1.MULTIMODE USER TERMINALS

- Terminal Design issues.
- Power consumption.
- System compatibility.

- **2. WIRELESS SYSTEM DISCOVERY**

  - Processing of the signals.
  - Different protocols access. .

3. WIRELESS SYSTEM SELECTION

- Proliferation complicates service selection .
- Resource optimization . .

1. TERMINAL MOBILITY

- Location management
- Handoff management .

2. NETWORK INFRASTRUCTURE

-Integrating the existing non-IP-based (voice) and IP-based (data)systems .

- Reliable wireless coverage.

4G Key challenges -System

QOS SUPPORT

- Guarantee QoS.
  - Accommodation of different qos requirements.
  - Time-sensitive or Multimedia applications .
  - Packet delay and packet dropping probabilities.

4.

SECURITY

- Schemes designed for specific services, (voice service) may not be applicable to 4G environments.
- The additional requirement for flexibility and mobility.
- Open protocols and standards.

The Focused Challenge

- In order to provide wireless services at any time and anywhere, terminal mobility is a must in 4G infrastructure
- Terminal mobility allows mobile clients to roam across geographical boundaries of the wireless networks
- There are two main issues in terminal mobility
  - location management
  - handoff management

Study of Middleware in Mobile Dialogue Control

Class : III yr- 5 SEM- CSE

Course Instructor : Dr. E. M. Malathy

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Session Objective

- To study the Middleware in Mobile Dialogue Control
- To understand the enabling communication, data management, and service coordination with mobile dialogue control

Session-outcome

- Students will be able to discover the characteristics of mobile computing applications including the major system components.

Middleware in Mobile Dialogue Control

- **Definition**:

Middleware is software that sits between the mobile OS/application layer and the network or hardware layer, enabling communication, data management, and service coordination.

Key Contributions:

- **Session Management**:
  - Maintains context across intermittent mobile sessions.
  - Ensures continuity even with mobile disconnections or handovers.
- **Message Coordination & Sequencing**:
  - Ensures correct order and delivery of messages.
  - Supports asynchronous communication between user interface and back-end services.
- **Service Abstraction**:
  - Abstracts complex back-end logic and offers unified API s.
  - Allows mobile apps to invoke services (like location, storage, AI processing) seamlessly.
- **Security and Authentication**:
  - Manages secure token exchange, identity verification, and encryption.
  - Controls dialogue access based on user roles or device profiles.

- **Context-Awareness**:
  - Middleware can adjust dialogue based on user location, device state, or preferences.
  - Enables adaptive interaction patterns (e.g., changing UI behavior based on bandwidth).

Examples:

- **Android Middleware Layer** (Binder, Content Providers)
- **Mobile Enterprise Application Platforms (MEAP)** like Kony, IBM Work-light
- **Middleware in IoT** for mobile-sensor integration (e.g., Edge X Foundry)

✓ 2. Gateways in Mobile Dialogue Control

Definition: A gateway is a node that acts as a translator, mediator, or bridge between different network types or protocols, often linking mobile devices to back-end systems or cloud.

Key Contributions:

- **Protocol Translation**:
  - Converts between mobile-specific protocols (e.g., HTTP/REST, MQTT) and enterprise protocols (e.g., SOAP, AMQP).
  - Ensures cross-platform dialogue (e.g., mobile-to-legacy system).
- **Data Filtering and Aggregation**:
  - Reduces unnecessary dialogue by pre-processing or aggregating data before forwarding.
  - Minimizes bandwidth use and improves dialogue relevance.

- **Quality of Service (QoS) Control**:
  - Ensures prioritized data flow (e.g., emergency alerts over routine sync).
  - Manages retries, buffering, and fallback mechanisms.
- **Firewall and Security Mediation**:
  - Acts as a security checkpoint to enforce policies on message exchanges.
  - Supports secure tunneling (VPN, TLS) for sensitive dialogues.

- **Edge Gateway Role**:
  - Performs edge computing tasks like caching or local AI inference.
  - Reduces round-trips and enhances responsiveness of dialogue.

Examples:

- **IoT Gateways** (e.g., AWS IoT Green-grass, Azure IoT Edge)
- **Mobile Network Gateways** (e.g., GGSN/PGW in 4G/5G)
- **API Gateways** (e.g., Kong, Apigee, AWS API Gateway)

Combined Role in Dialogue Control

Function	Middleware Role	Gateway Role
Session continuity	Tracks context and manages reconnections	Buffers and retries dialogue at network edges
Data transformation	Serializes/deserializes message formats	Translates between protocol formats
Security enforcement	Handles app-layer tokens and permissions	Manages secure data paths between networks
Load balancing	Routes requests to backend services	Routes messages across network paths
Real-time processing	Invokes edge/cloud services based on input	Executes edge computing to reduce latency

Summary

- **Middleware** manages **application-level dialogue logic**, such as session control, message sequencing, and API handling.
- **Gateways** ensure **network-level dialogue translation**, data optimization, and security bridging.
- Together, they orchestrate reliable, secure, and context-aware **dialogue control** that is essential for the seamless operation of mobile computing systems—especially in dynamic, multi-device environments like IoT, smart cities, and mobile cloud computing.

Study of **Wireless Application Protocol**

Class : III yr- 5 SEM- CSE

Course Instructor : Dr. E. M. Malathy

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Wireless Application Protocol

- What is WAP?
- WAP Architecture
- WAE (WML/WMLScript)
- WTA Framework
- WAP Push Services
- WAP Protocol Stack
- References and Resources

Mobile Applications - 1

- Vehicles
  - transmission of news, road condition etc
  - ad-hoc network with near vehicles to prevent accidents
- Emergencies
  - early transmission of patient data to the hospital
  - ad-hoc network in case of earthquakes, cyclones
  - military ...
- Traveling salesmen
  - direct access to central customer files
  - consistent databases for all agents

Mobile Applications - 1

- Web access
  - outdoor Internet access
  - intelligent travel guide with up-to-date location dependent information
- Information services
  - push: stock quotes; pull: nearest cash ATM
- Disconnected operations
  - file-system caching for off-line work
  - mobile agents, e.g., shopping
- Entertainment
  - games, etc

World Wide Web and Mobility

HTTP/HTML have not been designed for mobile applications/devices

- HTTP 1.0 characteristics
  - designed for large bandwidth, low delay
  - stateless, client/server, request/response communication
  - connection oriented, one connection per request
  - TCP 3-way handshake, DNS lookup overheads
  - big protocol headers, uncompressed content transfer
  - primitive caching (often disabled, dynamic objects)
  - security problems (using SSL/TLS with proxies)
- HTML characteristics
  - designed for computers with "high" performance, color high-resolution display, mouse, hard disk
  - typically, web pages optimized for design, not for communication; ignore end-system characteristics

System Support for Mobile WWW

- Enhanced browsers
  - client-aware support for mobility
- Proxies
  - Client proxy: pre-fetching, caching, off-line use
  - Network proxy: adaptive content transformation for connections
  - Client and network proxy
- Enhanced servers
  - server-aware support for mobility
  - serve the content in multiple ways, depending on client capabilities
- New protocols/languages
  - WAP/WML

Wireless Application Protocol (WAP)

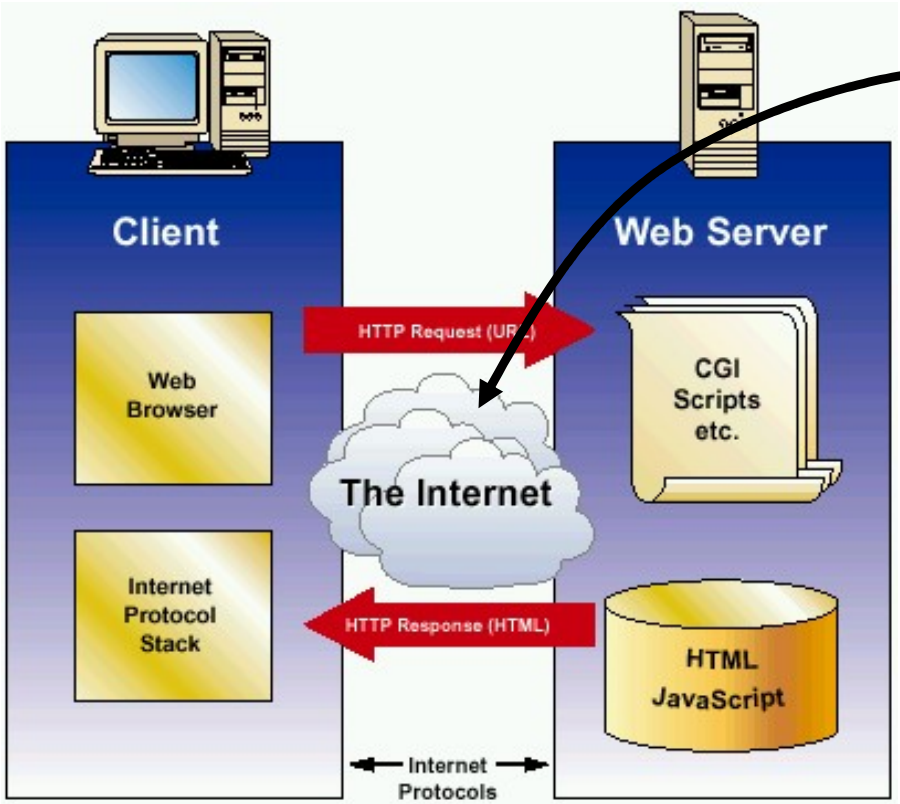
- Empowers mobile users with wireless devices to easily access and interact with information and services.
- A “standard” created by wireless and Internet companies to enable Internet access from a cellular phone

- wapforum.org:
  - co-founded by Ericsson, Motorola, Nokia, Phone.com
  - 450 members in 2000, comprise of Handset manufacturers, Wireless service providers, ISPs, Software companies in the wireless industry
  - Goals
    - deliver Internet services to mobile devices
    - enable applications to scale across a variety of transport options and device types
    - independence from wireless network standards
    - GSM, CDMA IS-95, TDMA IS-136, 3G systems (UMTS, W-CDMA)

WAP: Main Features

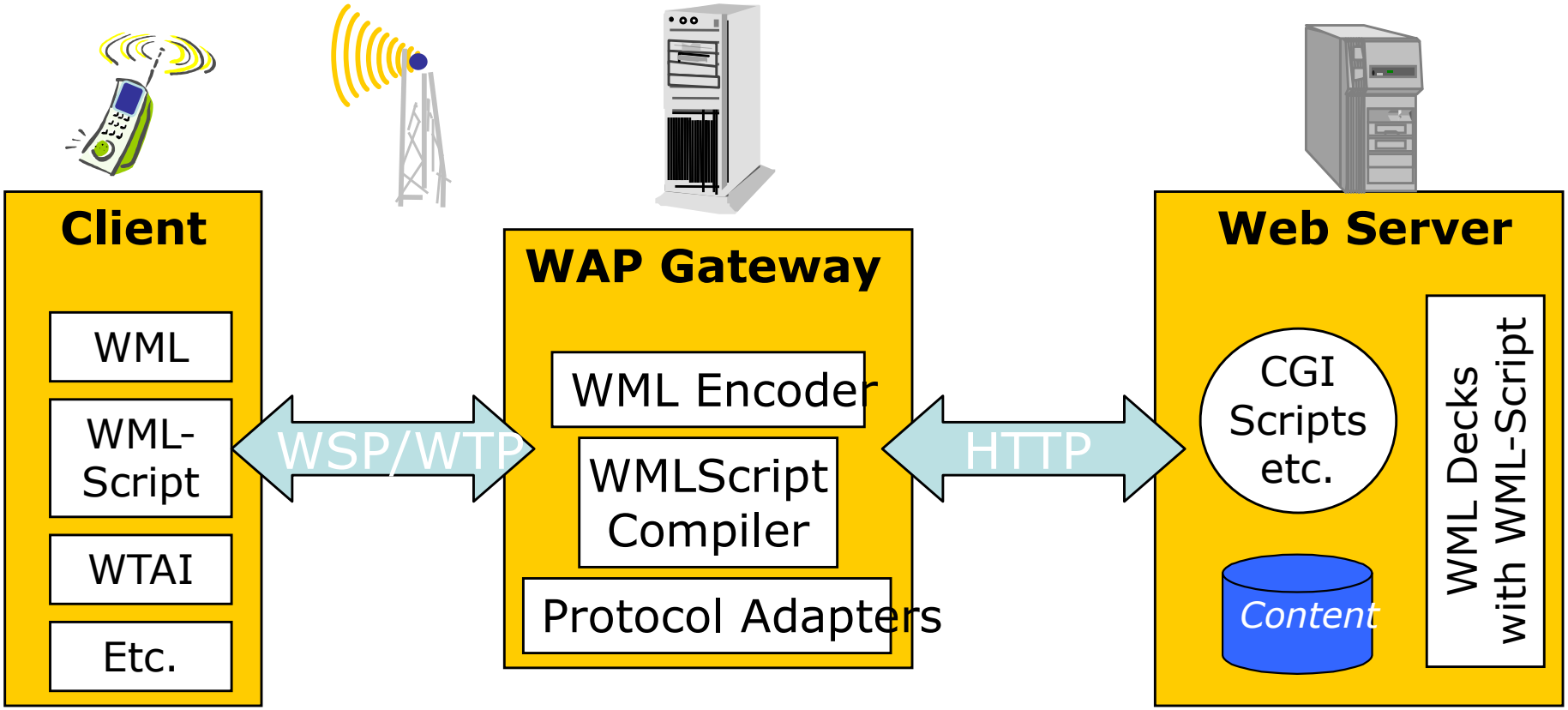
- Browser
  - "Micro browser", similar to existing web browsers
- Markup language
  - Similar to HTML, adapted to mobile devices
- Script language
  - Similar to Javascript, adapted to mobile devices
- Gateway
  - Transition from wireless to wired world
- Server
  - "Wap/Origin server", similar to existing web servers
- Protocol layers
  - Transport layer, security layer, session layer etc.
- Telephony application interface
  - Access to telephony functions

Internet Model



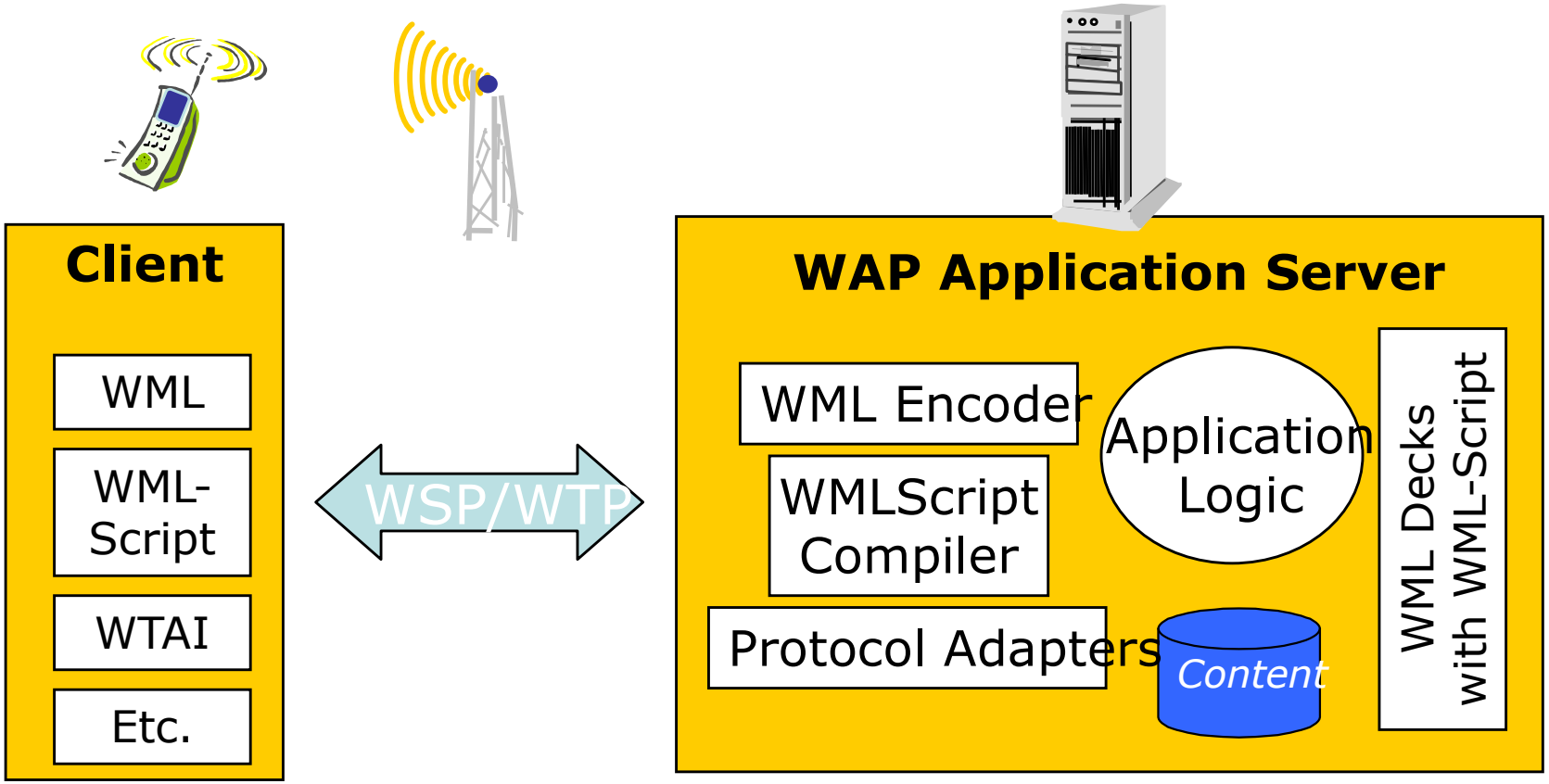
HTML
HTTP
TLS/SSL
TCP/IP

WAP Architecture



Source: WAP Forum

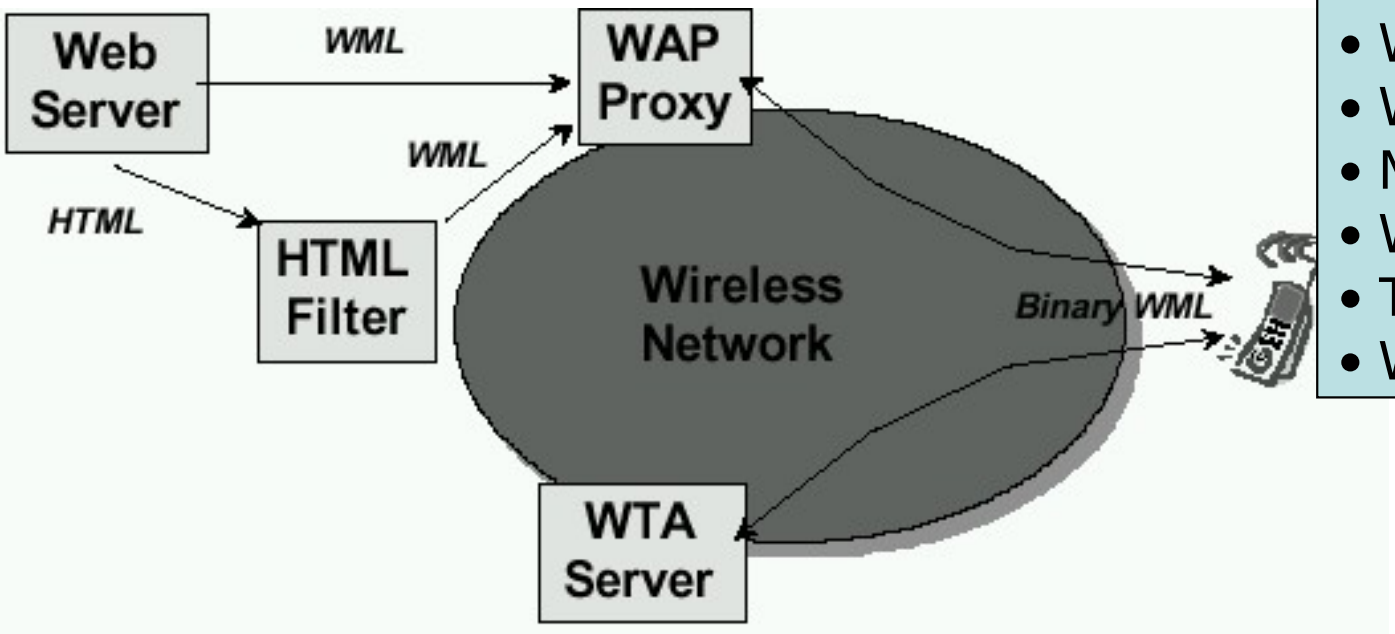
WAP Application Server



Source: WAP Forum

WAP Architecture

- Another look

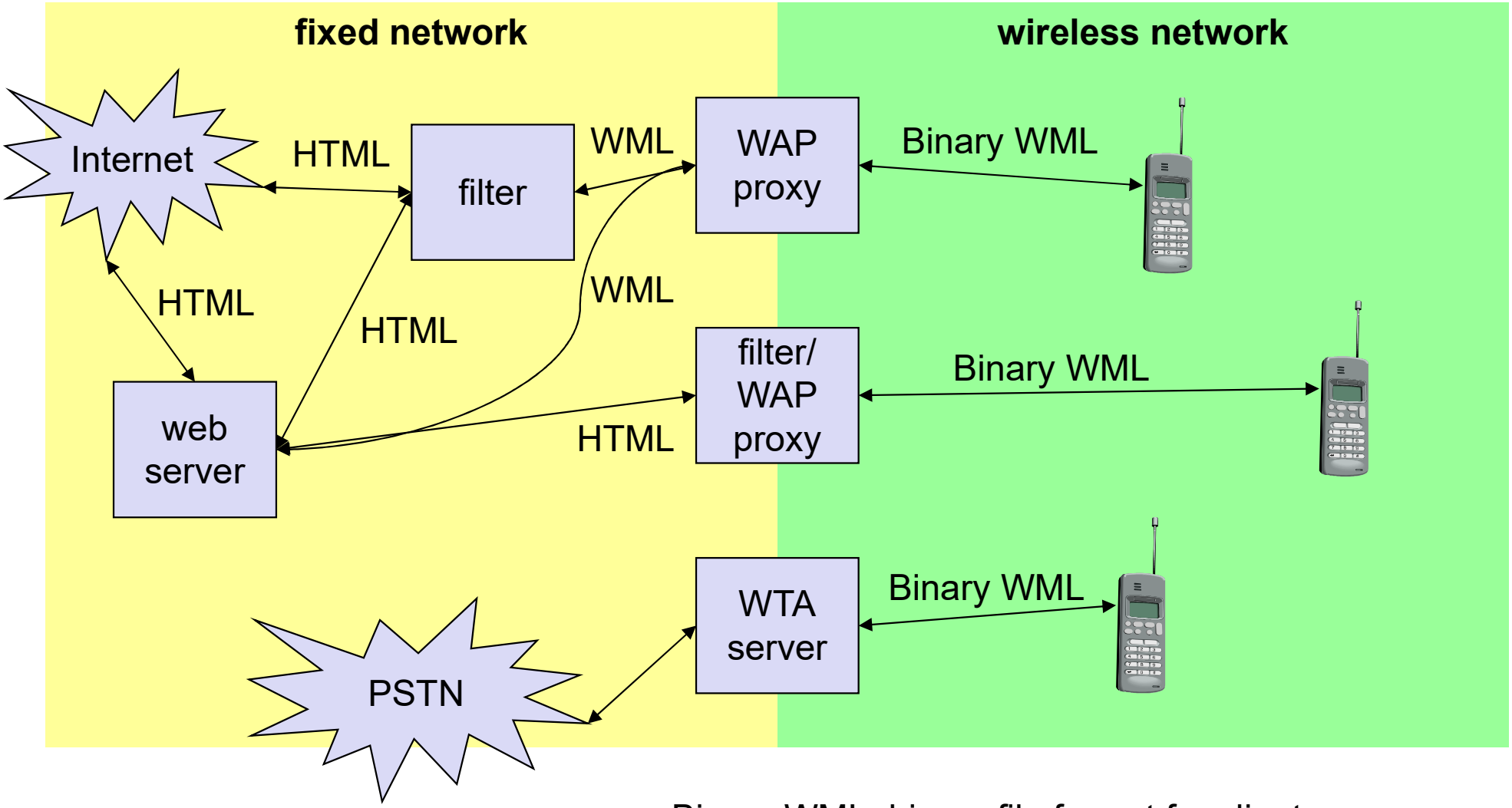


Key Components

- Origin/Web Server
- WAP Gateway/Proxy
- WAP Protocol Stack
- Micro Browser
- WML/WML Script
- Transcoders
- WTA

Source: WAP Forum

WAP: Network Elements



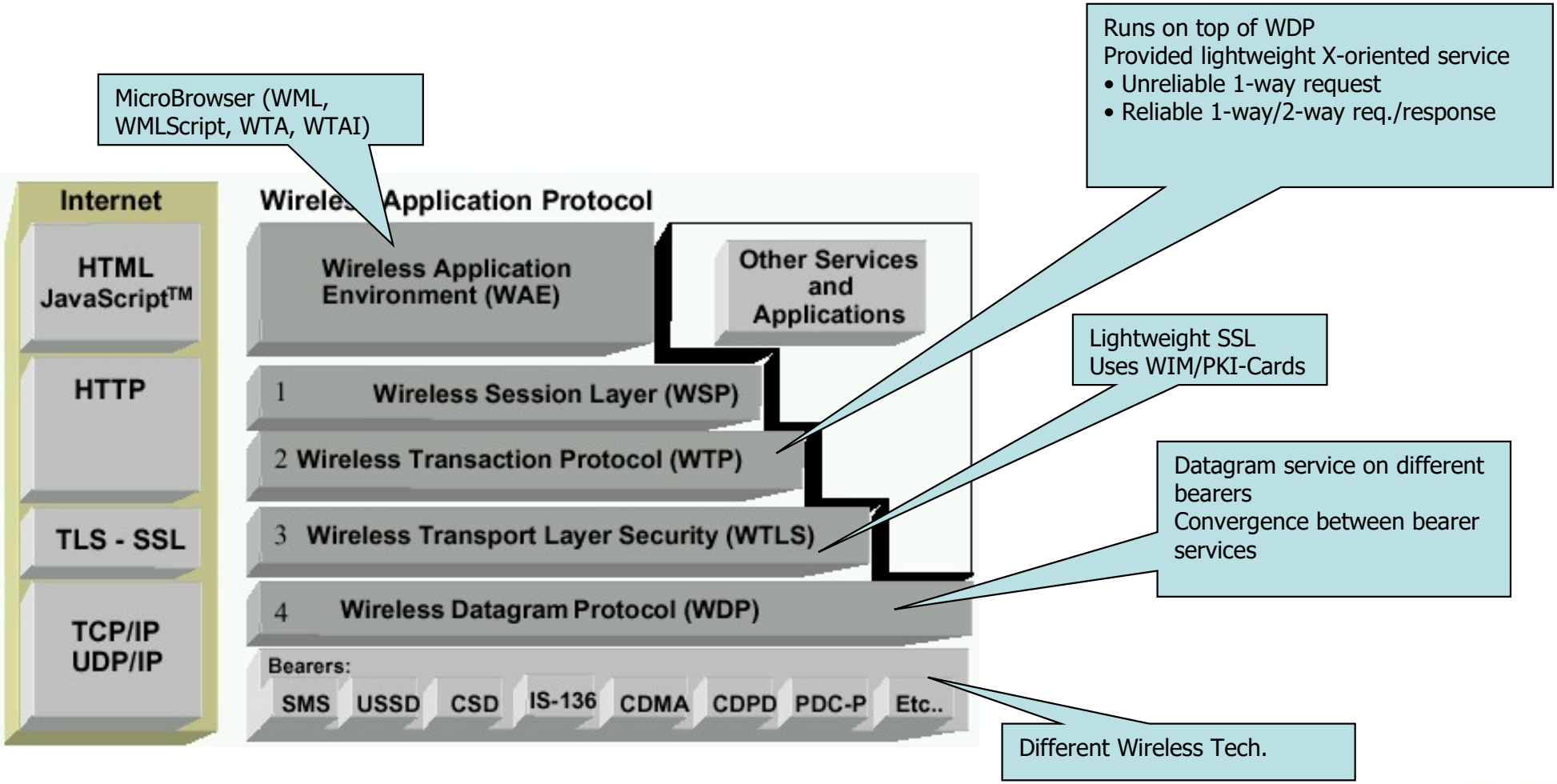
Binary WML: binary file format for clients

Source: Schiller

WAP Specifies

- Wireless Application Environment
  - WML Microbrowser
  - WMLScript Virtual Machine
  - WMLScript Standard Library
  - Wireless Telephony Application Interface (WTAI)
  - WAP content types
- Wireless Protocol Stack
  - Wireless Session Protocol (WSP)
  - Wireless Transport Layer Security (WTLS)
  - Wireless Transaction Protocol (WTP)
  - Wireless Datagram Protocol (WDP)
  - Wireless network interface definitions

WAP Stack



Source: WAP Forum

WAP Stack

- WAE (Wireless Application Environment):
  - Architecture: application model, browser, gateway, server
  - WML: XML-Syntax, based on card stacks, variables, ...
  - WTA: telephone services, such as call control, phone book etc.
- WSP (Wireless Session Protocol):
  - Provides HTTP 1.1 functionality
  - Supports session management, security, etc.
- WTP (Wireless Transaction Protocol):
  - Provides reliable message transfer mechanisms
  - Based on ideas from TCP/RPC
- WTLS (Wireless Transport Layer Security):
  - Provides data integrity, privacy, authentication functions
  - Based on ideas from TLS/SSL
- WDP (Wireless Datagram Protocol):
  - Provides transport layer functions
  - Based on ideas from UDP

Content encoding, optimized for low-bandwidth channels, simple devices

WHY WAP?

- Wireless networks and phones
  - have specific needs and requirements
  - not addressed by existing Internet technologies
- WAP
  - Enables any data transport
    - TCP/IP, UDP/IP, GUTS (IS-135/6), SMS, or USSD.
  - Optimizes the content and air-link protocols
  - Utilizes plain Web HTTP 1.1 servers
    - leverages existing development methodologies
    - utilizes standard Internet markup language technology (XML)
    - all WML content is accessed via HTTP 1.1 requests
  - WML UI components map well onto existing mobile phone user interfaces
    - no re-education of the end-users
    - leveraging market penetration of mobile devices
  - Several modular entities together form a fully compliant Internet entity

Why is HTTP/HTML not enough?

Big pipe - small pipe syndrome

Internet

HTTP/HTML

```
<HTML>
<HEAD>
<TITLE>NNN Interactive</TITLE>
<META HTTP-EQUIV="Refresh" CONTENT="1800,
URL=/index.html">
</HEAD>
<BODY BGCOLOR="#FFFFFF"
BACKGROUND="/images/9607/bgbar5.gif" LINK="#0A3990"
ALINK="#FF0000" VLINK="#FF0000" TEXT="000000"
ONLOAD="if(parent.frames.length!=0)top.location='ht
tp://nnn.com';">
<A NAME="#top"></A>
<TABLE WIDTH=599 BORDER="0">
<TR ALIGN=LEFT>
<TD WIDTH=117 VALIGN=TOP ALIGN=LEFT>
```

```
<HTML>
<HEAD>
<TITLE
>NNN
Intera
ctive<
/TITLE
>
<META
HTTP-
EQUIV=
"Refre
sh"
CONTEN
T="180
0,
URL=/i
ndex.h
tml">
```

Wireless network

WAP

```
<WML>
<CARD>
<DO TYPE="ACCEPT">
<GO URL="/submit?Name=$N"/>
</DO>
Enter name:
<INPUT TYPE="TEXT" KEY="N"/>
</CARD>
</WML>
```

Content encoding

```
010011
010011
110110
010011
011011
011101
010010
011010
```

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Source: WAP Forum

WAP: “Killer” Applications

- Location-based services
  - Real-time traffic reporting, Event/restaurant recommendation
- Enterprise solutions
  - Email access, Database access, “global” intranet access
  - Information updates “pushed” to WAP devices
- Financial services
  - Banking, Bill-paying, Stock trading, Funds transfers
- Travel services
  - Schedules and rescheduling, Reservations
- Gaming and Entertainment
  - Online, real-time, multi-player games
  - Downloadable horoscopes, cartoons, quotes, advice
- M-Commerce
  - Shopping on the go
  - Instant comparison shopping
  - Location-based special offers and sales

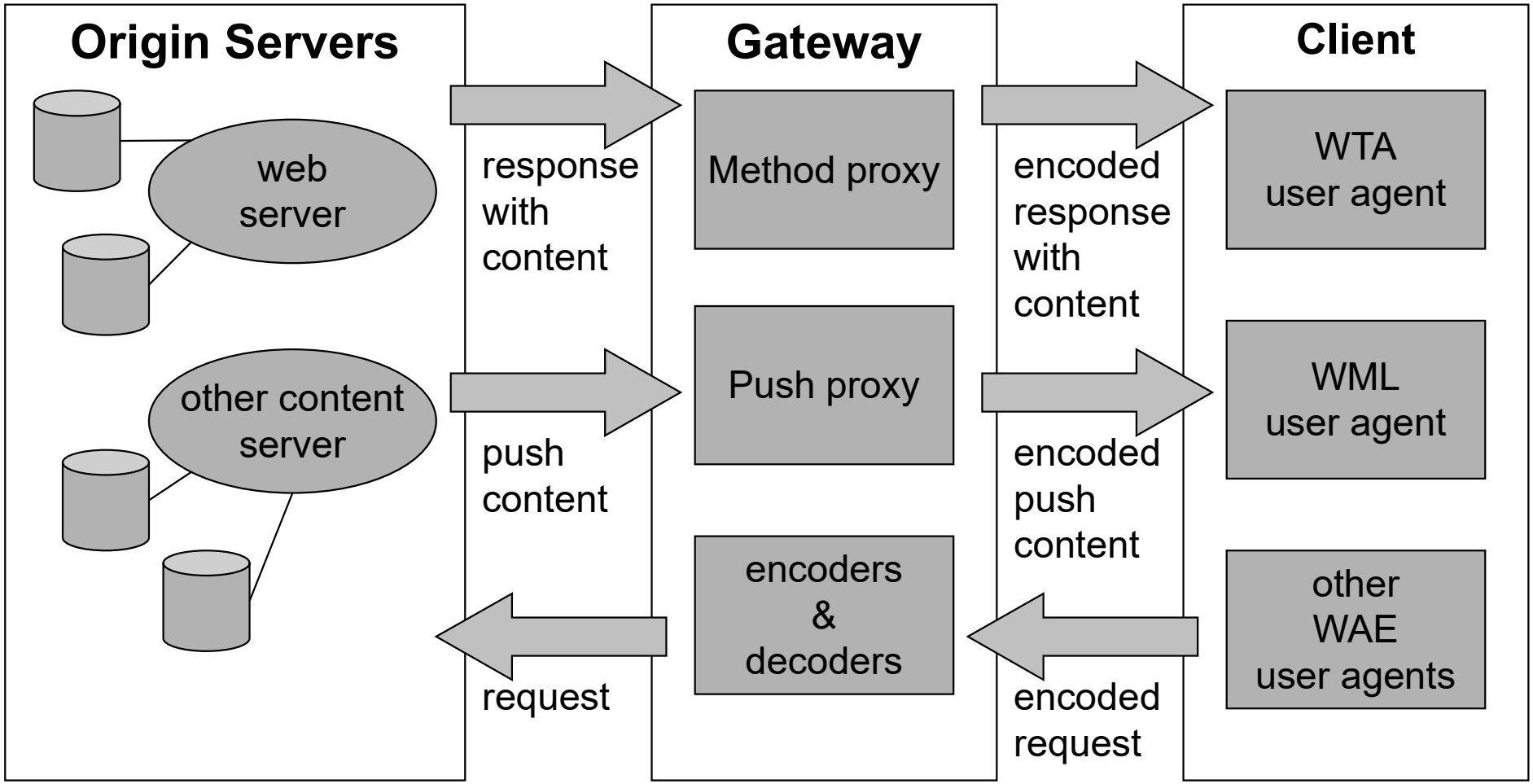
Wireless Application Environment (WAE)

- Goals
  - device and network independent application environment
  - for low-bandwidth, wireless devices
  - considerations of slow links, limited memory, low computing power, small display, simple user interface (compared to desktops)
  - integrated Internet/WWW programming model
  - high interoperability

WAE Components

- Architecture
  - Application model, Microbrowser, Gateway, Server
- User Agents
  - WML/WTA/Others
  - content formats: vCard, vCalendar, Wireless Bitmap, WML, ...
- WML
  - XML-Syntax, based on card stacks, variables, ...
- WMLScript
  - procedural, loops, conditions, ... (similar to JavaScript)
- WTA
  - telephone services, such as call control, text messages, phone book, ... (accessible from WML/WMLScript)
- Proxy (Method/Push)

WAE: Logical Model

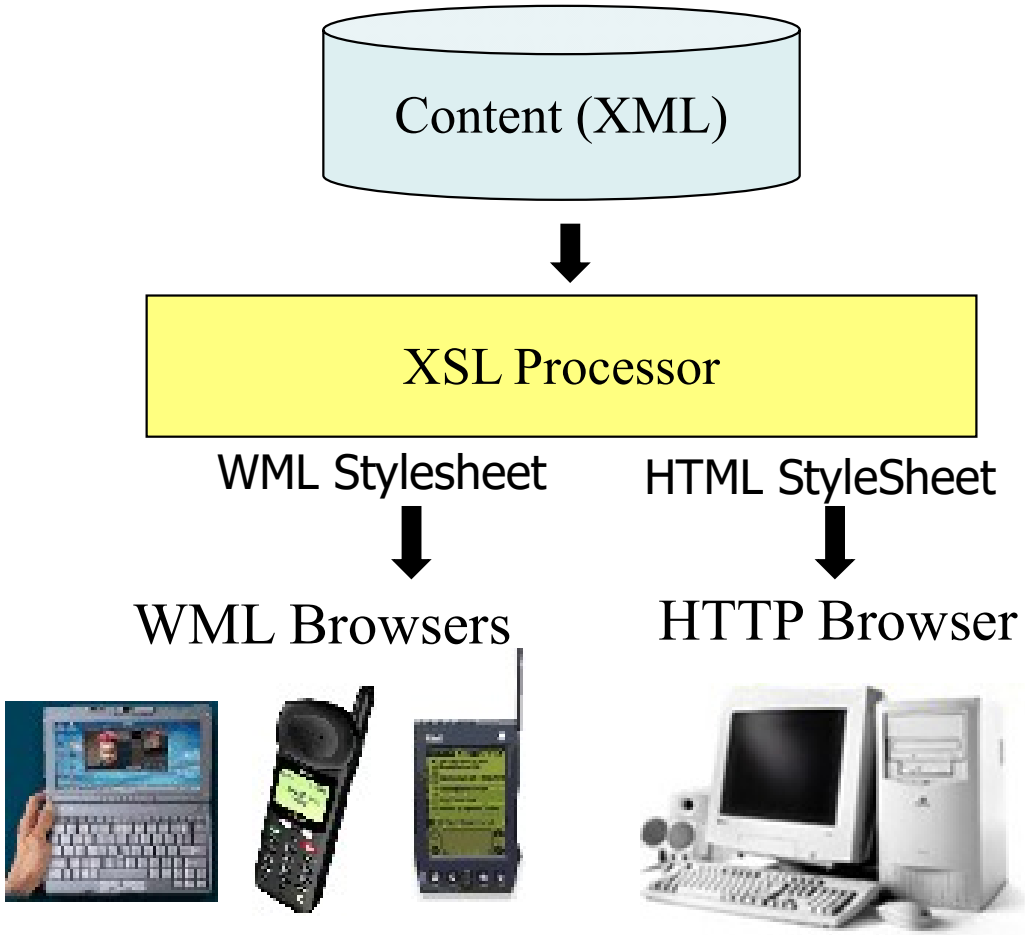


WAP Microbrowser

- Optimized for wireless devices
- Minimal RAM, ROM, Display, CPU and keys
- Provides consistent service UI across devices
- Provides Internet compatibility
- Enables wide array of available content and applications

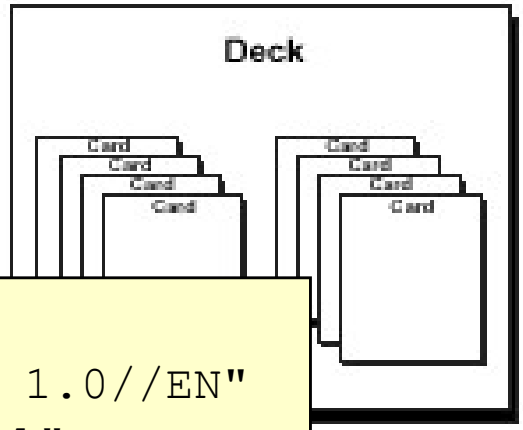
WML: Wireless Markup Language

- Tag-based browsing language:
  - Screen management (text, images)
  - Data input (text, selection lists, etc.)
  - Hyperlinks & navigation support
- Takes into account limited display, navigation capabilities of devices
- XML-based language
  - describes only intent of interaction in an abstract manner
  - presentation depends upon device capabilities
- Cards and Decks
  - document consists of many cards
  - User interactions are split into cards
  - Explicit navigation between cards
  - cards are grouped to decks
  - deck is similar to HTML page, unit of content transmission



WML

- The basic unit is a **card**. Cards are grouped together into **Decks** Document ~ Deck (unit of transfer)
- All decks must contain
  - Document prologue
    - XML & document type declaration
  - <WML> element
    - Must contain one or more cards



WML File Structure

```
<?xml version="1.0"?>
<!DOCTYPE WML PUBLIC "-//WAPFORUM//DTD WML 1.0//EN"
        "http://www.wapforum.org/DTD/wml.xml">

<WML>
    ...
</WML>
```

WML (other features)

- Setting card styles to create forms
- Using variables to cache user data
- Using card intrinsic events to trigger transparent tasks
- Using timers
- Securing WML decks
- Bookmarking decks

WMLScript

- Complement to WML
  - Derived from JavaScript™
- Provides general scripting capabilities
  - Procedural logic, loops, conditionals, etc.
  - Optimized for small-memory, small-cpu devices

- Features
  - local user interaction, validity check of user input
  - access to device facilities (phone call, address book etc.)
  - extensions to the device software
    - configure device, download new functionality after deployment

- Bytecode-based virtual machine
  - Stack-oriented design, ROM-able
  - Designed for simple, low-impact implementation
- WMLScript compiler resides in the network

WMLScript Libraries

- Lang - VM constants, general-purpose math functionality, etc.
- String - string processing functions
- URL - URL processing
- Browser - WML browser interface
- Dialog - simple user interface
- Float - floating point functions

WMLScript Example

Functions

Variables

Programming Constructs

```
function currencyConvertor(currency, exchRate)
{
    return currency*exchangeRate;
}

function myDay(sunShines) {
    var myDay;
    if (sunShines) {
        myDay = "Good";
    } else {
        myDay = "Not so good";
    };
    return myDay;
}
```

Source: WAP Forum

Wireless Telephony Application (WTA)

- Collection of telephony specific extensions
  - designed primarily for network operators

- Example
  - calling a number (WML)
    `wtai://wp/mc;07216086415`
  - calling a number (WMLScript)
    `WTAPublic.makeCall("07216086415");`

WTA Features

- Extension of basic WAE application model
  - network model for interaction
    - client requests to server
    - event signaling: server can push content to the client
  - event handling
    - table indicating how to react on certain events from the network
    - client may now be able to handle unknown events
  - telephony functions
    - some application on the client may access telephony functions
- WTAI includes:
  - Call control
  - Network text messaging
  - Phone book interface
  - Event processing
- Security model: segregation
  - Separate WTA browser
  - Separate WTA port

Push Access Protocol

- Based on request/response model
- Push initiator is the client
- Push proxy is the server
- Initiator uses HTTP POST to send push message to proxy
- Initiator sends control information as an XML document, and content for mobile (as WML)
- Proxy sends XML entity in response indicating submission status
- Initiator can
  - cancel previous push
  - query status of push
  - query status/capabilities of device

Push Proxy Gateway

- WAP stack (communication with mobile device)
- TCP/IP stack (communication with Internet push initiator)
- Proxy layer does
  - control information parsing
  - content transformation
  - session management
  - client capabilities
  - store and forward
  - prioritization
  - address resolution
  - management function

WAE Summary

- WML
  - analogous to HTML (optimized for wireless)
  - event based, microbrowser user agent
- WMLScript
  - analogous to JavaScript
  - features of compiler in the network
- WTA
  - WTAI: different access rights for different applications/agents
  - WTA User Agent (analogy with operating systems)
    - Context – Activation Record
    - Channel – Interrupt Handler
    - Resource – Shared routines invoked by interrupt handlers
    - Repository – Library of interrupt handlers
  - feature of dynamically pushing the interrupt handler before the event
- Push
  - no analogy in Internet

WAP Gateway Summary

- Encoders
  - translate between binary (WML) and text (HTML/WML)
- Filters
  - transcoding between WML (wireless) and HTML (wired)
- Method Proxy
  - similar to standard proxy services
  - WAP stack on wireless interface and TCP/IP stack on Internet interface
- Push Proxy
  - Push Access Protocol with Internet Push

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WAP Servers Summary

- Origin Server
  - Web server with HTML/WML contents
  - Runs TCP/IP stack, needs PAP protocol for push, no end-to-end security
- WAP Server
  - Serves WML content
  - Runs WAP stack, uses OTA protocol for push, end-to-end security possible
- WTA Server
  - Specialized for telephony applications (runs WAP stack, uses push extensively)
  - Client initiated (make call “hyperlink” from a Yellow pages service)
  - Server intiated (incoming call from a Voice mail service)

SECURITY IN MOBILE NETWORKS

Mobile Computing Environment

- Vulnerable to failures, intrusion, and eavesdropping.
- Adhoc mobile systems has everything moving (hosts, base-stations, routers/agents, subnets, intranet).
- Need survivability from intentional and unintentional attacks.

Deficiency in Mobile IP Authentication

- Authentication is through a home agent (HA).
  - If HA is out of service, mobile host will be homeless and not be able to communicate.

Deficiency in Mobile IP Key Management

- Data packets are encrypted before sending, and decrypted after receiving.
- Requires exchange of secret keys and public keys between sender and receiver.
- Mobile IP does not provide multi-cast session key management.

Objective

- To provide uninterrupted secure service to the mobile hosts when base station moves or fails.

Mobile IP Entities

- **Mobile Host (MH)** – which can change its point of attachment to the internet from one link to another.
- **Home Agent (HA)** – router on MH's home network which tunnels datagrams (packets of data) to MH when it is away from home.
- **Foreign Agent (FA)** – router on MH's visited network which provides routing services to the MH while registered.

Hardware Characteristics

- **Media** – Wireless media are inherently less secure.
- **Low power and limited computing resource** – motivation for making security an optional feature.
- **Bandwidth** – typically orders of magnitude less than wired bandwidth (motivation for reducing the overhead of the security scheme).

Application Characteristics

- **Location Privacy** – protecting the identity of the communicating entities (ex: Military Networks)
- **Mobility** – implies frequent upon handoffs
- **Secure Multicast** – one transmitter and many listeners (ex: Classrooms)

Fundamental Security Services

- **Authentication**
  - Provides assurance of a host's identity.
  - Provides a means to counter masquerade and replay attacks.
  - Can be applied to several aspects of multicast (ex: registration process).

Fundamental Security Services

- **Integrity**
  - Provides assurance that traffic is not altered during the transmission.
  - Lack of integrity services in IP can lead to spoofing attacks.
  - More crucial for applications involving key management than voice applications (easily detected).

Fundamental Security Services

- **Confidentiality**
  - Provides assurance that only authorized entities can decode and read the data.
  - Typically, encryption is used to achieve this.
  - Encryption can be applied at several layers of the protocol stack (ex: inherent in RTP, ESP for IP datagrams).

Fundamental Security Services

- **Other Services**
- **Non-repudiation –** recipient can prove that sender did sent the message in case sender denies it.
- **Access Control –** ensures that only authorized parties can access the resources.

Problem Description

- To ensure security and theft of resources (like bandwidth), all the packets originating inside the network should be authenticated.

- Typically, a Mobile Host sends a packet to its Home Agent along with the authentication information.

Problem Description (continued)

- If the Authentication is successful, Home Agent forwards the packet. Otherwise, packet is dropped.

Disadvantages of Typical Setup

- ~~Home Agent becomes a single point of~~ failure.
- Home agent becomes an attractive spot for attackers.
- Not scalable – large number of hosts overload the Home agent.

Research Goals

- ~~Eliminate the single point of failure.~~
- Distribute the load and enhance scalability and survivability of the system.
- Failures – transparent to applications.
- Easy to implement, no manual setup.

Traditional Approaches

- Using a Proxy Server (or Backup) that takes up the responsibilities of the Base Station

Disadvantages

- Manual updating of the routing tables of the hosts necessary.
- Time consuming and hence smooth provision of service is not possible.

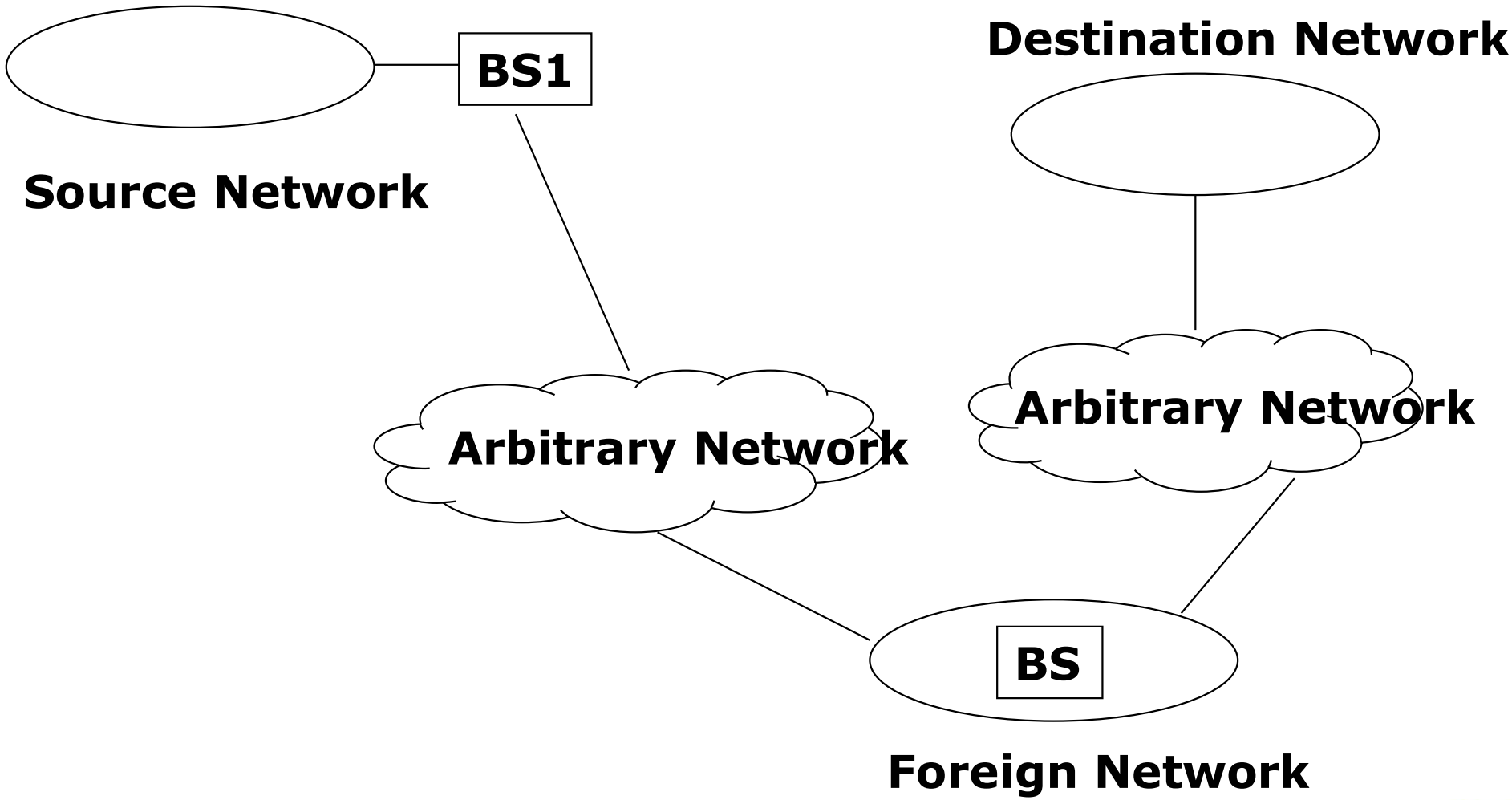
Traditional Approaches (continued)

- Using a Second Base Station that forwards the packets to the actual Home Agent, using Mobile IP, which is now at a Foreign Network.

Disadvantages

- Communication Delays introduced makes this solution impractical.
- Introduces additional security threats as the packets now traverse long paths through Internet.

Proxy-Based Solution



Disadvantages

- Introduces additional security threats.
- Additional communication delays.
- Not transparent to applications.
- Manual set up – error prone.

Mobile Security Services

- Two schemes to solve the problem.
  - Virtual Home Agent
  - Hierarchical Authentication
- They differ in the architecture and the responsibilities that the Mobile Hosts and Base Stations (Agents) hold.

Authentication Using Virtual Home Agent

Entities in the proposed scheme

- Virtual Home Agent (VHA) is an abstract entity identified by a network address.
- Master Home Agent (MHA) is the physical entity that carries out the responsibilities of the VHA.

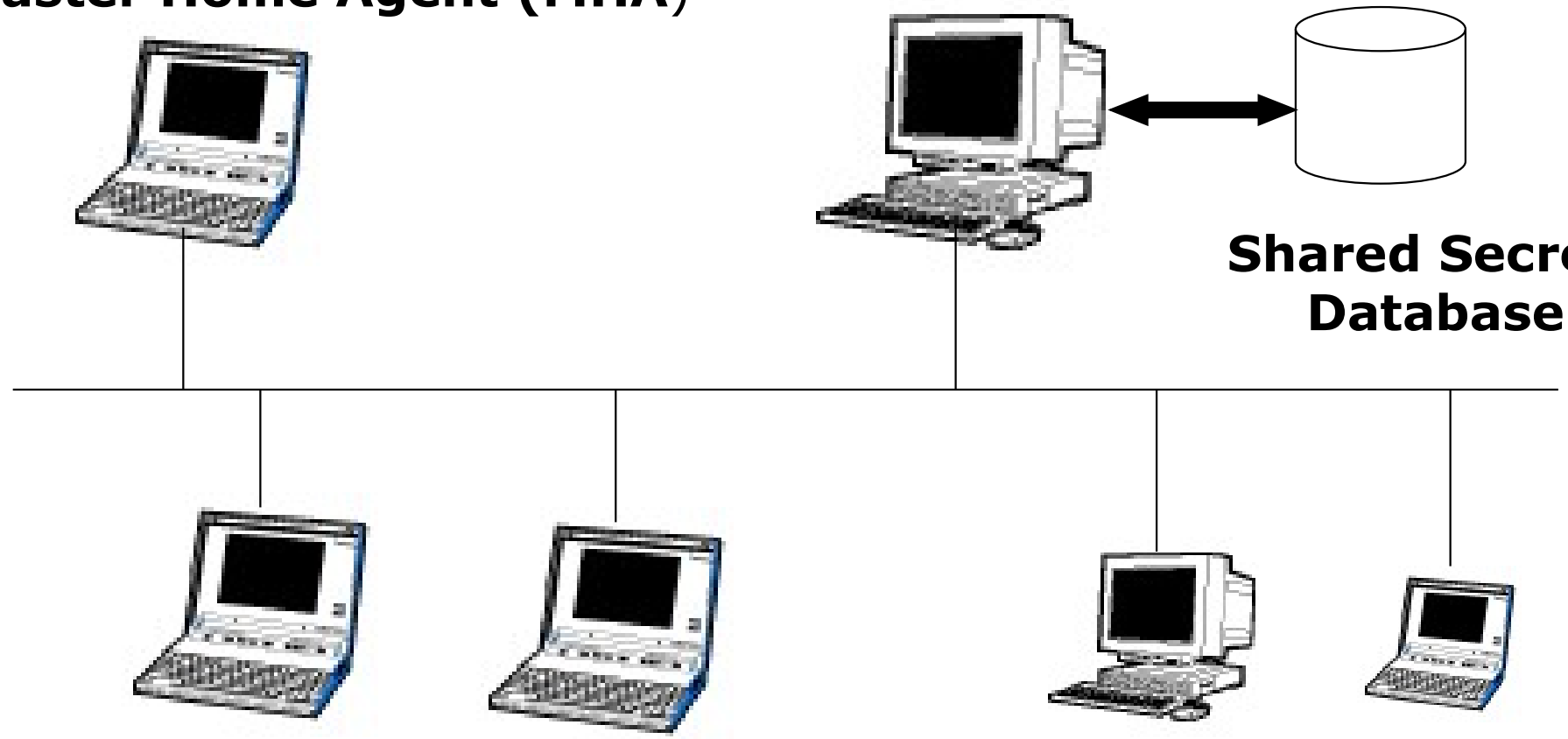
Authentication Using Virtual Home Agent

- Backup Home Agent (BHA) is the entity that backs up a VHA. When MHA fails, BHA having the highest priority becomes MHA.
- Shared Secrets Database Server is the entity that manages and processes the queries on the secret database.

Virtual Home Agent Scheme

VHA ID = IP ADDRI
Master Home Agent (MHA)

Database Server



Hierarchical Authentication Scheme

Key Priority depends on several factors and computed as cumulative sum of weighted priorities of each factors:

Example Factors:

- Communication Delays
- Processing Speed of the Agents
- Key Usage
- Life Time of the Key

Hierarchical Authentication Scheme

- Hosts detect the Home Agent's failure or mobility when the Home Agent does not send an acknowledgement for a request.
- When the failure is detected, host reduces the priority of the current key and picks up the highest priority key to be used from now onwards.

VHA Scheme	Hierarchical Scheme
• Flat structure • Host has only one key • Failure is transparent to the user	• Tree structure • Number of keys depend on height of the tree. • Hosts should be aware of the failure of BS as which key to be used depends on the base station serving it.
• No Priority is assigned to the keys	• Each key has priority, the key with the highest priority is used for authentication.

Future Work

- Quantifying the priorities for each factor and computing the overall key priority as a weighted function of all these factors.
- Designing a adaptable database replication and partitioning scheme for secret key database that increases the system performance.
- Simulation of these approaches and obtaining performance statistics.

UNIT 1
Mobile computing through internet

CS23512 - FUNDAMENTALS OF MOBILE COMPUTING

Class: V SEM CSE

Course Instructor: Dr. E. M. Malathy

Course Outcome

- CO1: Discover the characteristics of mobile computing applications including the major system components
- CO2: To explore the working model and characteristics of mobile computing
- CO3:To propose solutions with comparisons for problems related to mobile computing system through system investigations
- CO4:To identify the use of wireless technologies in appropriate applications and Quantum programming frameworks for mobile applications
- CO5:Develop a mobile application using mobile technologies and Emerging research level understanding at the intersection of mobile and quantum computing

- What is Mobile Computing?
- Importance in modern-day communication and productivity
- Overview of the session:
  - Internet-based mobile computing
  - Quantum-inspired algorithms

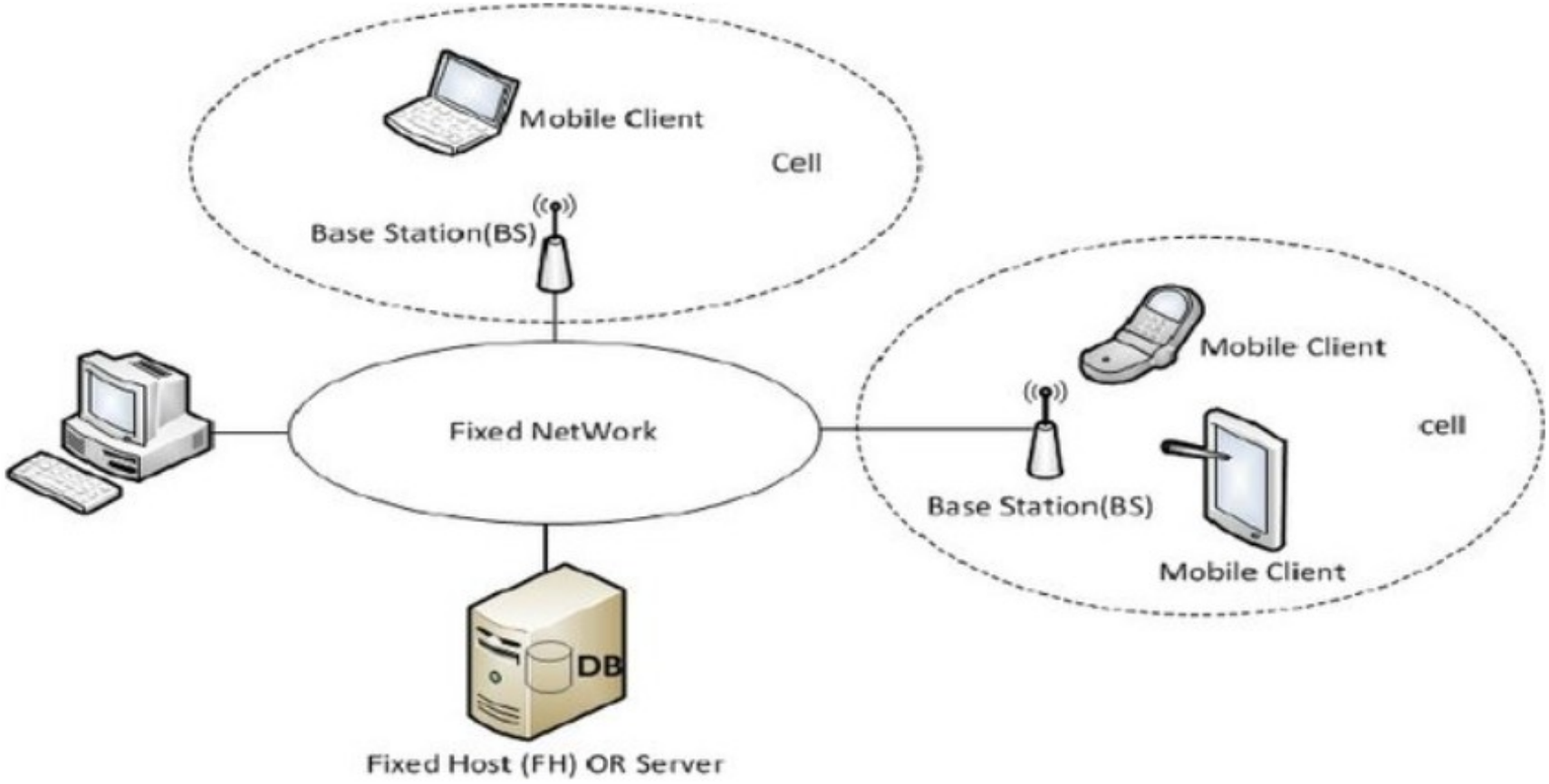
Mobile Computing through Internet

- Definition: Accessing data/services via mobile devices through the internet
- Examples: Cloud storage, mobile apps, IoT control, web-based services
- Key Components:
  - Wireless communication (WiFi, LTE, 5G)
  - Mobile devices (smartphones, tablets)
  - Cloud infrastructure (AWS, Azure)

Architecture for Internet-Based Mobile Computing

- **Client Side:**
  - Lightweight applications
  - Responsive UIs
- **Server Side:**
  - REST APIs
  - Cloud-based processing and storage
- **Middleware:**
  - Authentication (OAuth, JWT)
  - Synchronization and session management

Client-Server-Mobile Architecture



Challenges in Mobile Computing

- Network latency and inconsistency
- Power consumption
- Data security and privacy
- Limited resources (CPU, memory)

Introduction to Quantum-Inspired Algorithms

- What are quantum-inspired algorithms?
  - Use principles like superposition, entanglement metaphorically
  - Designed to optimize complex systems using classical hardware
- Why use them in mobile computing?
  - Improve optimization and efficiency
  - Handle large datasets and decision problems

Applications of Quantum-Inspired Algorithms in Mobile

- **Routing optimization** for navigation apps
- **Battery management** for longer device life
- **Data compression** and encryption
- **Resource scheduling** in cloud-mobile architecture

Comparative Benefits

Feature	Traditional Algorithm	Quantum-Inspired
Speed	Moderate	High (on optimization)
Resources	Higher	Lower (more efficient)
Complexity	Simple to medium	Higher but better scalability

https://www.youtube.com/watch?v=JhHMJCUmq28

- https://www.youtube.com/feed/downloads

Session Summary

- Internet enables mobile computing powerfully, but faces challenges
- Quantum-inspired algorithms offer innovative ways to improve mobile app performance
- Future looks promising with more hybrid computing models

Study of Mobile Computing Architecture

Class : III yr- 5 SEM- CSE

Course Instructor : Dr. E. M. Malathy

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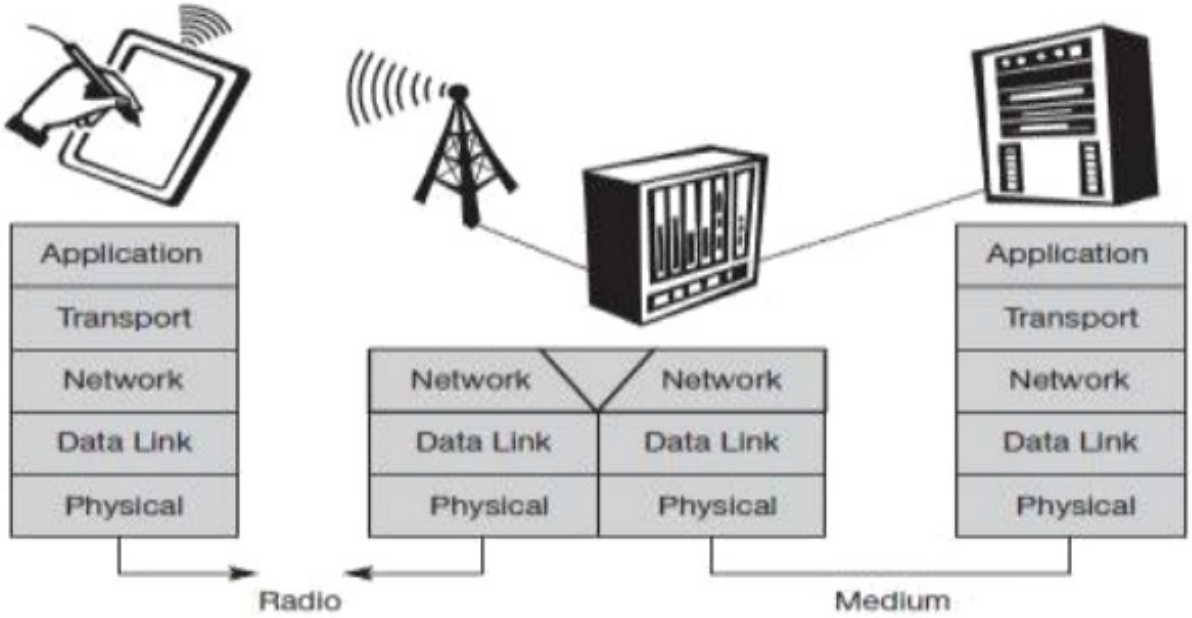
A simplified reference model

The figure shows the **protocol stack** implemented in the system according to the reference model. **End-systems**, such as the PDA and computer in the example, need a full protocol stack comprising the application layer, transport layer, network layer, data link layer, and physical layer. Applications on the end-systems communicate with each other using the lower layer services. **Intermediate systems**, such as the interworking unit, do not necessarily need all of the layers.


Application
Transport
Network
Data Link
Physical
Network
Network
Data Link
Data Link
Physical
Physical
Application
Transport
Network
Data Link
Physical
Radio
Medium

A Simplified Reference Model

- **Physical layer**: This is the lowest layer in a communication system and is responsible for the conversion of a stream of bits into signals that can be transmitted on the sender side. The physical layer of the receiver then transforms the signals back into a bit stream. For wireless communication, the physical layer is responsible for frequency selection, generation of the carrier frequency, signal detection (although heavy interference may disturb the signal), modulation of data onto a carrier frequency and (depending on the transmission scheme) encryption.

- **Data link layer**: The main tasks of this layer include accessing the medium, multiplexing of different data streams, correction of transmission errors, and synchronization (i.e., detection of a data frame). Altogether, the data link layer is responsible for a reliable point-to-point connection between two devices or a point-to-multipoint connection between one sender and several receivers.

- **Network layer**: This third layer is responsible for routing packets through a network or establishing a connection between two entities over many other intermediate systems. Important functions are addressing, routing, device location, and handover between different networks.

- **Transport layer**: This layer is used in the reference model to establish an end-to-end connection

- **Application layer**: Finally, the applications (complemented by additional layers that can support applications) are situated on top of all transmission oriented layers. Functions are service location, support for multimedia applications, adaptive applications that can handle the large variations in transmission characteristics, and wireless access to the world-wide web using a portable device.

Summary

- Mobile computing architecture typically involves a three-tier structure: a presentation tier for user interfaces, an application tier for business logic and transactions, and a data tier for database access. Middleware connects these tiers, managing functions like networking and security. This architecture enables mobility, meaning users can access services and applications from various locations and devices.